Important Information

Thank you for choosing Freenove products!

Getting Started

If you have not yet downloaded the zip file associated with this kit, please do so now and unzip it.
https://github.com/Freenove/Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi

Please refer to the specific tutorial based on the product model you purchased.

If your model is FNK0108P/Q/R/U/V/W (shown below), follow the **Pro_Version_Tutorial.pdf** (the current documentation) to assemble the product.



If your model is FNK0108A/B/C/H/K/L (as shown in the figure below), please follow the "**Base_Version_Tutorial.pdf**" file located in the same folder for assembly.



Get Support and Offer Input

Freenove provides free and responsive product and technical support, including but not limited to:

- Product quality issues
- Product use and build issues
- Questions regarding the technology employed in our products for learning and education
- Your input and opinions are always welcome
- We also encourage your ideas and suggestions for new products and product improvements

For any of the above, you may send us an email to:

support@freenove.com

Safety and Precautions

Please follow the following safety precautions when using or storing this product:

- Keep this product out of the reach of children under 6 years old.
- This product should be **used only when there is adult supervision present** as young children lack necessary judgment regarding safety and the consequences of product misuse.
- This product contains small parts and parts, which are sharp. This product contains electrically conductive parts. **Use caution with electrically conductive parts near or around power supplies, batteries and powered (live) circuits.**
- When the product is turned ON, activated or tested, some parts will move or rotate. **To avoid injuries to hands and fingers keep them away from any moving parts!**
- It is possible that an improperly connected or shorted circuit may cause overheating. **Should this happen, immediately disconnect the power supply or remove the batteries and do not touch anything until it cools down!** When everything is safe and cool, review the product tutorial to identify the cause.
- Only operate the product in accordance with the instructions and guidelines of this tutorial, otherwise parts may be damaged or you could be injured.
- Store the product in a cool dry place and avoid exposing the product to direct sunlight.
- After use, always turn the power OFF and remove or unplug the batteries before storing.

About Freenove

Freenove provides open source electronic products and services worldwide.

Freenove is committed to assist customers in their education of robotics, programming and electronic circuits so that they may transform their creative ideas into prototypes and new and innovative products. To this end, our services include but are not limited to:

- Educational and Entertaining Project Kits for Robots, Smart Cars and Drones
- Educational Kits to Learn Robotic Software Systems for Arduino, Raspberry Pi and micro: bit
- Electronic Component Assortments, Electronic Modules and Specialized Tools
- **Product Development and Customization Services**

You can find more about Freenove and get our latest news and updates through our website:

http://www.freenove.com

sale@freenove.com

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Preface

Welcome to use the Freenove Computer Case Kit Mini for Raspberry Pi



This product is exclusively designed for the RPi 5, a popular single-board computer (SBC). Follow this tutorial, you can install a sleek, multi-functional computer case kit for your Raspberry Pi 5 (RPi 5).

The RPi 5 delivers highly powerful performance with a CPU frequency of up to 2.4GHz. However, this increased power consumption also leads to significant heat generation. Without an appropriate heatsink, thermal throttling may severely limit the Raspberry Pi 5's performance. Additionally, the removal of the 3.5mm audio jack in the RPi 5 has caused inconvenience for many users.

In this context, our case kit came into being. It not only addresses these issues, but also introduces advanced features.

If you encounter any issues, feel free to contact us for prompt and free technical support.

support@freenove.com

Key Features

This product integrates a wide range of features and supports multiple optional expansions, as detailed below:

- **Standard Features:**
  - Built-in dual 4Ω 3W speakers for audio output.
  - Equipped with a 3.5mm JACK audio interface.
  - Applying a tower cooler for more efficient heat dissipation.
  - Built-in two ARGB 4010 cooling fans for lower overall case temperature and enhanced lighting effects
  - Classic case design.
  - Power button with indicator light.
  - Featuring a 1220 RTC battery holder.
  - Ports are uniformly arranged on the back of the case, including USB, Ethernet, UART, HDMI, 3.5mm JACK, SD card slot, and power interface for easy connection and management
- **Optional Features:**
  - Supports MVMe M.2 slot expansion in 2230/2242/2260/2280 specifications; optional configurations of 1/2/4 slots to meet different storage needs
  - Optional 128GB SSD for basic storage requirements

Product Variants

This product is currently available in several variants, with the primary distinction being **the number of NVMe slots**. Models FNK0108U, FNK0108V, and FNK0108W are additionally equipped with one NVMe M.2 SSD.

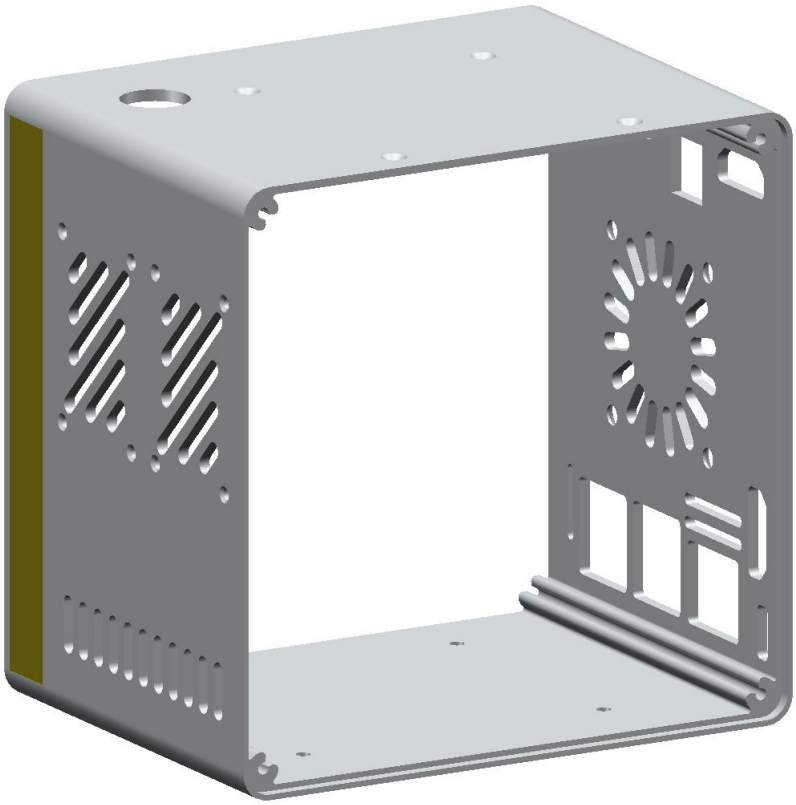
Models	Number of NVMe Slot	NVMe M.2 SSD x1
FNK0108P	1	✗
FNK0108Q	2	✗
FNK0108R	4	✗
FNK0108U	1	✓
FNK0108V	2	✓
FNK0108W	4	✓

List

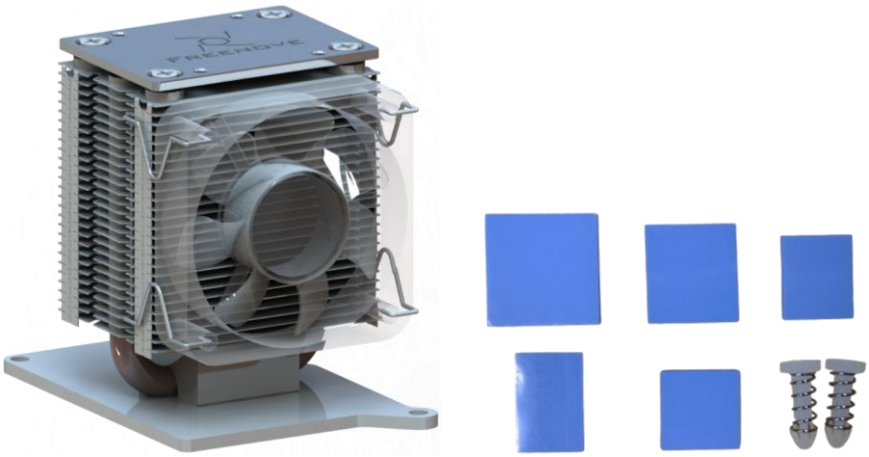
Before getting started, please check the part list. If any component is missing from your kit, do not start assembly; instead, please email support@freenove.com to get the missing parts.

Metal Parts

Metal case x1



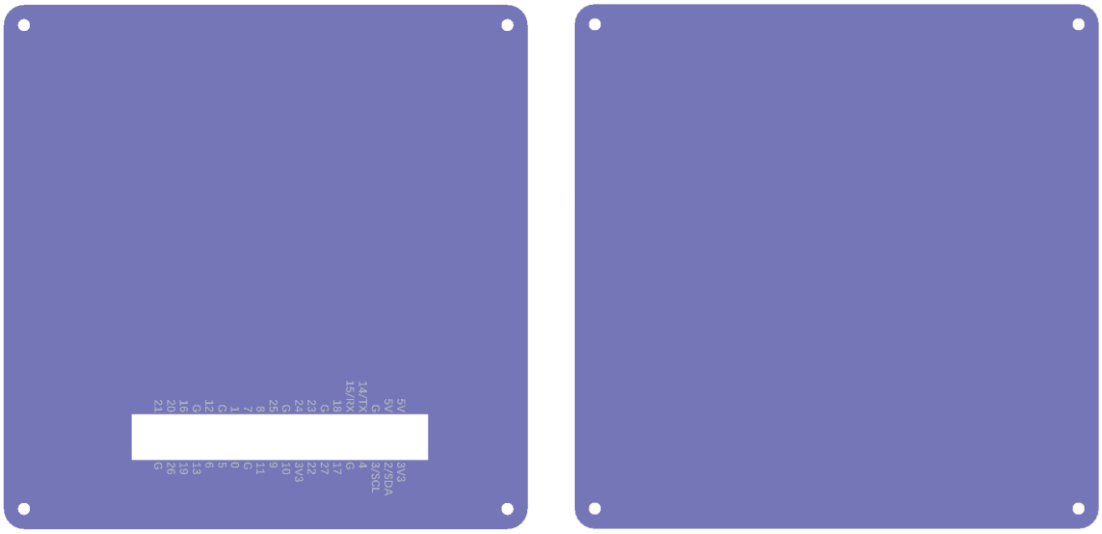
Tower Cooler x1
Thermal Pad x5
Nylon Standoff x2



Acrylic Parts

Note: Please tear off the protective films from the acrylic parts before use.

Acrylic Side Plate Set x1



Speaker Acrylic Pad x2



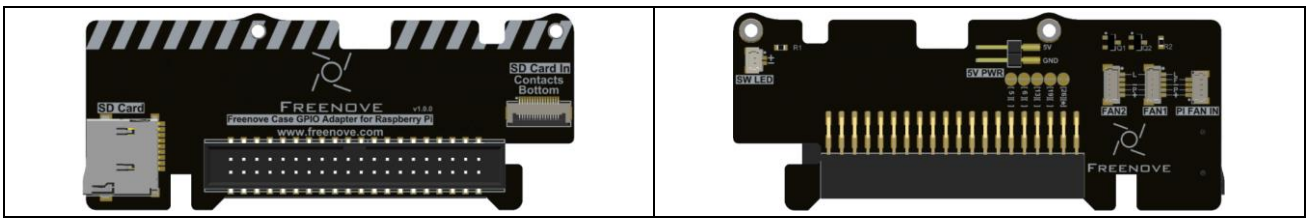
Machinery Parts

All fasteners come in a larger bag, please open it and check whether they are complete.

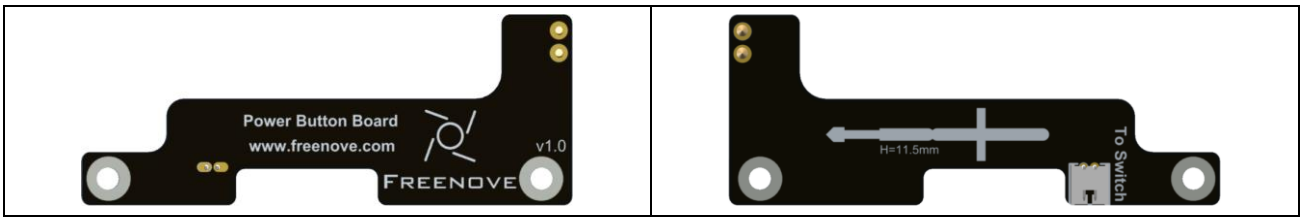
		
		

Electronic Parts

Freenove Case GPIO Adapter for Raspberry Pi



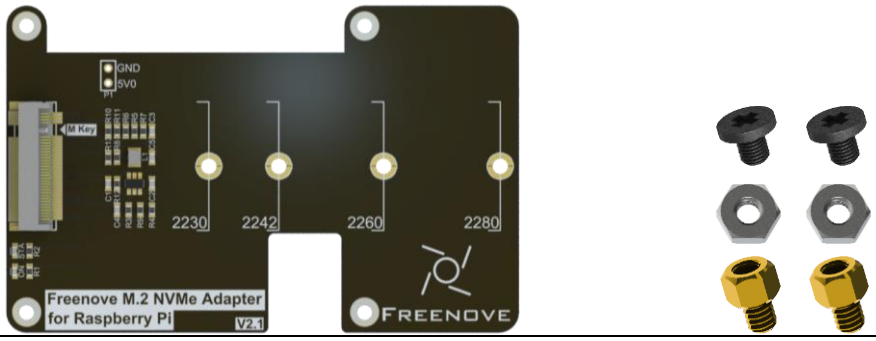
Freenove Power Button Board for Raspberry Pi



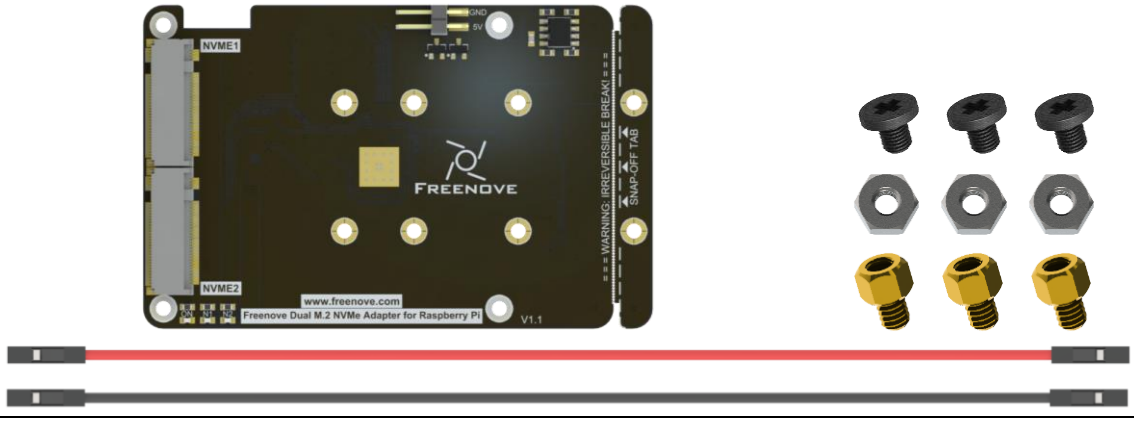
Freenove M.2 NVMe Adapter Series for Raspberry Pi

Note: The components included in the NVMe Adapter Combo Pack vary by product version. Please verify that the contents match your model before installation.

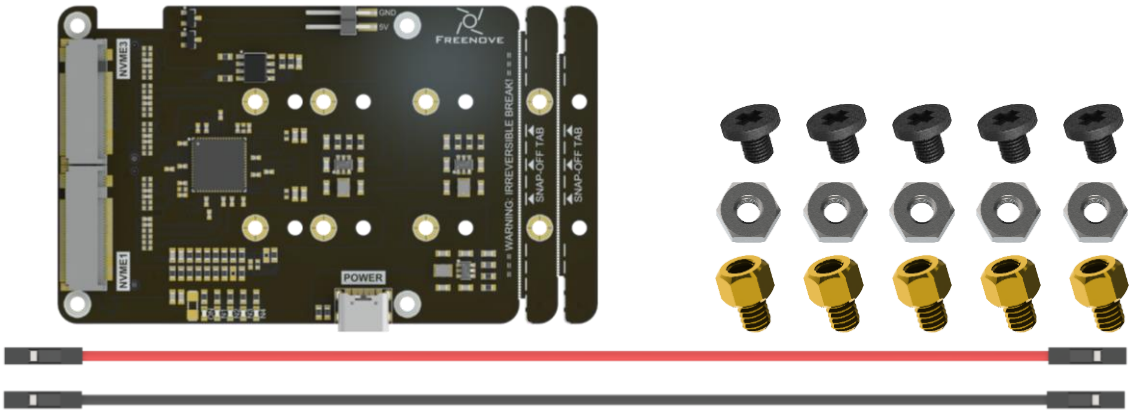
Freenove M.2 Nvme Adapter for Raspberry Pi Combo Pack x1 (Only for FNK0108P/U)



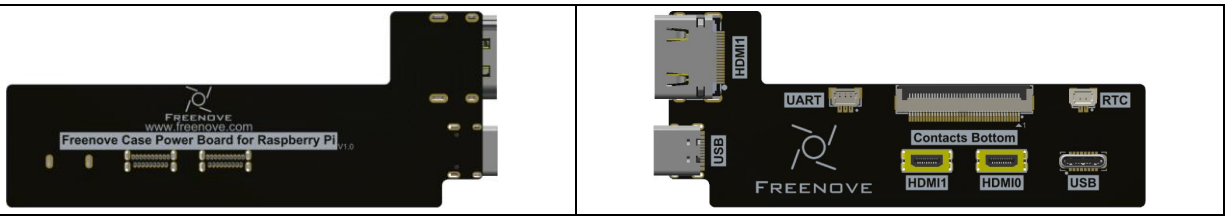
Freenove Dual M.2 Nvme Adapter for Raspberry Pi Combo Pack x1 (Only for FNK0108Q/V)



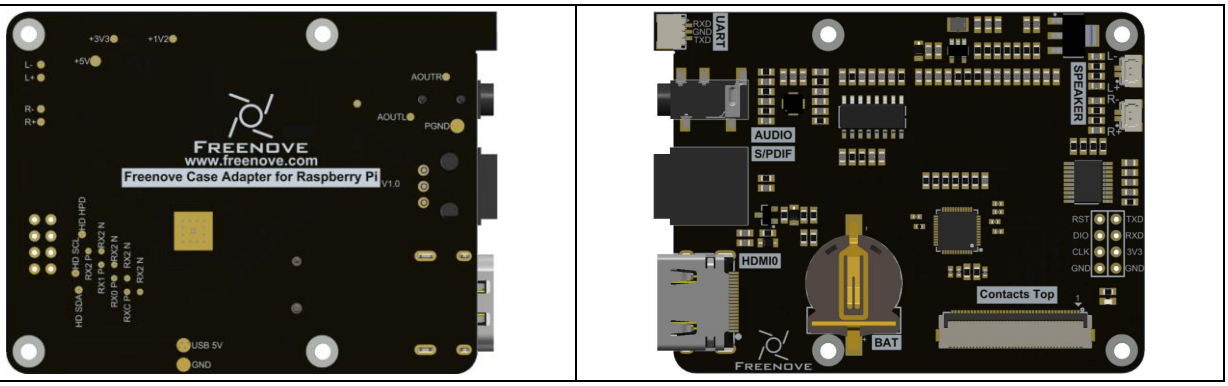
Freenove Quad M.2 Nvme Adapter for Raspberry Pi Combo Pack x1 (Only for FNK0108R/W)



Freenove Case Power Board for Raspberry Pi



Freenove Case Adapter for Raspberry Pi



Electronic Modules

ARGB Fan x1



Speaker x2



NVMe SSD x1 (Only FNK0108U/V/W)



Wires

SH1.0mm_2P Same-Direction Cable 5cm x1
SH1.0mm_3P Same-Direction Cable 5cm x1
SH1.0mm_4P Same-Direction Cable 5cm x1



SH1.0mm 2-Pin to 2.8mm Quick-Disconnect Terminal Cable (Red-Black), 7cm, ×1
1.25mm 2-Pin to 2.8mm Quick-Disconnect Terminal Cable (Yellow-Yellow), 7cm, ×1



86mm HDMI FPC Cable x1



SD Card to 0.5mm-16P FPC cable x1



PCIe FPC cable x1

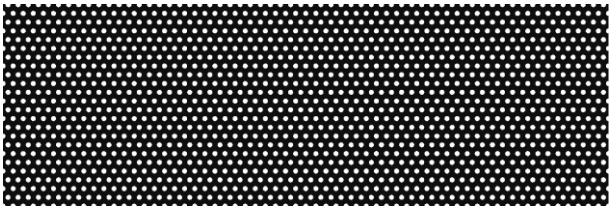
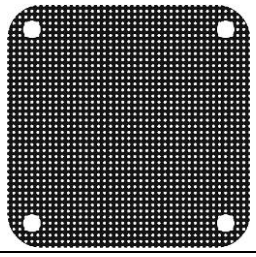
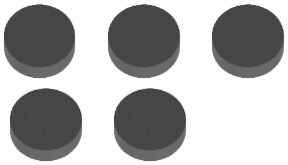


Tools

Screwdriver Bit Holder ×1
Hex Shank Phillips #2 Bit ×1
Hex Shank Phillips #0 Bit ×1



Others

12mm LED Power Button x1 Black Sealing Gasket x1 M12 Nut x1 	Air Inlet Dust Filter x1 
Fan Dust Filter x1 	Round Black Non-Slip Foot Pad x5 

Rquired but NOT Contained Parts

Raspberry Pi 5 x 1 
27W Power Adapter x 1 (or a power adapter compatible with Raspberry Pi official one that can output 5.1V/5A) 

Micro SD Card (TF Card), Card Reader x 1



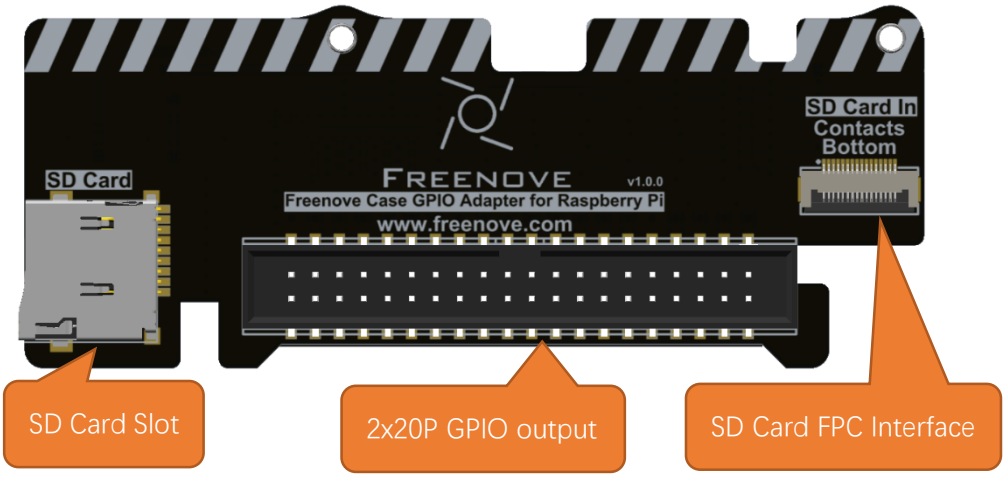
Before getting started, please check the part list. If any component is missing from your kit, do not start assembly; instead, please email our support team at **support@freenove.com** to get the missing parts.

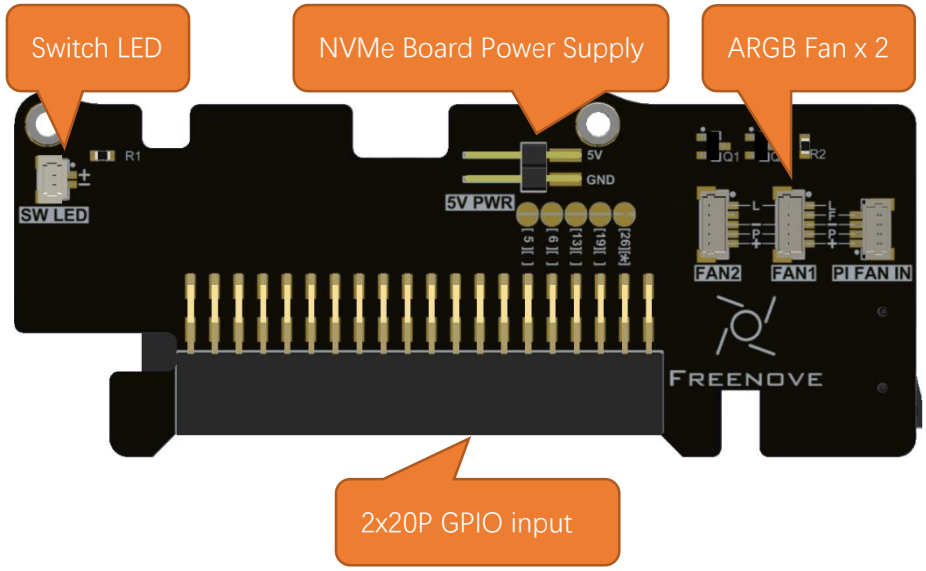
Chapter 1 Introduction to Main Components

In this chapter, we will introduce the main components of this case and their functionalities.

1.1 Introduction to Freenove Case GPIO Adapter for Raspberry Pi

In this tutorial, we name this model as **GPIO Board**.





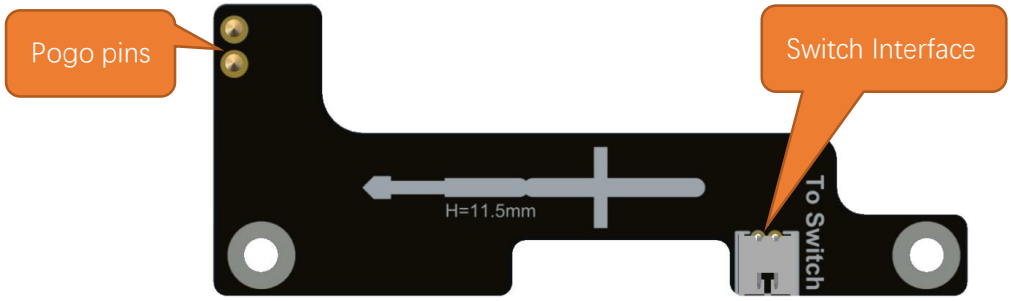
The GPIO Board serves to externally expose the Raspberry Pi 5's GPIO pins and SD card slot, allowing convenient connection of external peripherals and SD card access.

Its integrated features also include:

- Two ARGB fan headers for independent control of the tower cooler fan and the case fan (Multiple RGB LED interface controls are available.).
- LED control for the case power switch, providing a clear visual indication of the Raspberry Pi 5's power state (on/off).
- An auxiliary power connector for the NVMe Board.

1.2 Introduction to Freenove Power Button Board for Raspberry Pi

In this tutorial, we name this model as **Switch Board**.



The Switch Board connects to the J2 interface of the Raspberry Pi 5 via pogo pins, enabling the use of a custom power button. A 12mm power button with an indicator light is included, offering identical functionality to the onboard power button of the Raspberry Pi 5. For detailed specifications and usage of the power button, please refer to the Raspberry Pi official documentation:
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#power-button

1.3 Introduction to Freenove M.2 NVMe Adapter Series for Raspberry Pi

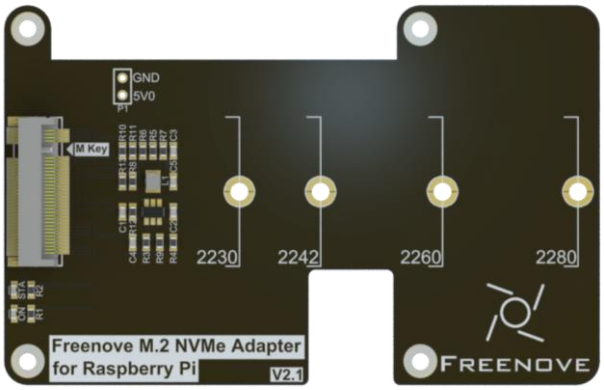
The Raspberry Pi 5 includes a PCIe x1 slot that is certified for PCIe Gen 2.0, providing a theoretical maximum throughput of 5GT/sec, which roughly translates to 500MB/sec for read and write operations. Although this slot is not officially certified for PCIe Gen 3.0, it is possible to force the use of Gen 3.0 for potentially higher speeds.

The PCIe consortium states that the speed of PCIe Gen 3.0 x1 is up to 8GT/sec, which translates to approximately 985MB/sec.
https://en.wikipedia.org/wiki/PCI_Express#Comparison_table
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#pcie-gen-3-0

SSDs generally provide significantly faster read and write speeds compared to SD cards and USB drives, which can notably elevate the user experience when operating the Raspberry Pi 5.

1.3.1 Freenove M.2 NVMe Adapter for Raspberry Pi

In this tutorial, we name this model as **NVMe Adapter Board**.

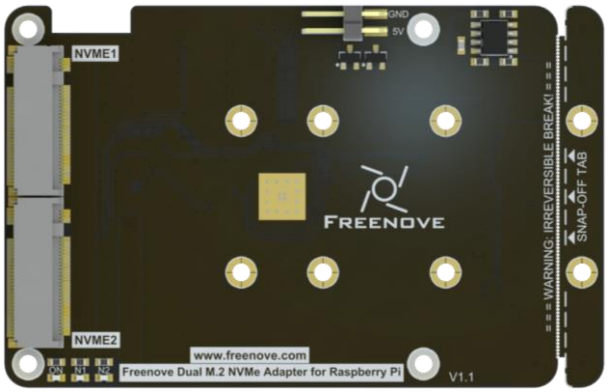


Here are its key features:

- **Interface Type**: M.2 with M-Key
- **Supported Protocol**: NVMe
- **PCIe Channel**: PCIe 3.0 x1(Compatible with PCIe 2.0)
- **Compatible Sizes**: 2230, 2242, 2260, 2280
- **Power Supply**: 3.3V, up to 3A (maximum)
- **Indicator Lights**: Includes both power and SSD status LEDs.

1.3.2 Freenove Dual M.2 NVMe Adapter for Raspberry Pi

This model has four NVMe SSD interface, supporting up to 2 NVMe SSDs to run simultaneously. In this tutorial, we name this model as **Dual-NVMe Adapter Board**.

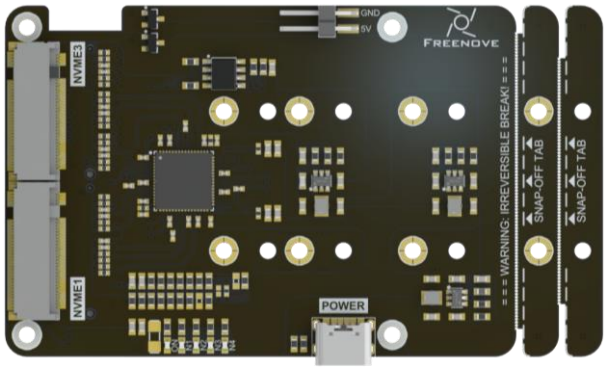


Here are its key features:

- **Interface Type**: 2x M.2 with M-Key
- **Supported Protocol**: NVMe
- **PCIe Channel**: PCIe 2.0 x2
- **Compatible Sizes**: 2230, 2242, 2260, 2280
- **Power Supply**: 3.3V, up to 3A (maximum)
- **Indicator Lights**: Includes both power and x2 SSD status LEDs.

1.3.3 Freenove Quad M.2 NVMe Adapter for Raspberry Pi

This model has four NVMe SSD interface, supporting up to 4 NVMe SSDs to run simultaneously. In this tutorial, we name this model as **Quad-NVMe Adapter Board**.

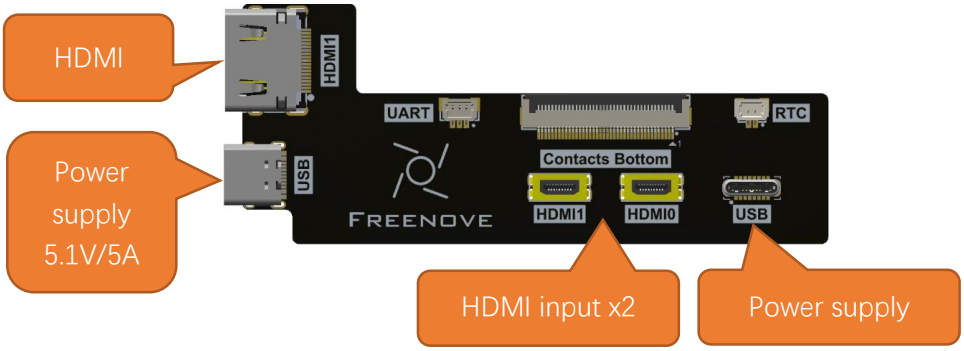


Here are its key features:

- **Interface Type**: 4x M.2 with M-Key
- **Supported Protocol**: NVMe
- **PCIe Channel**: PCIe 2.0 x4
- **Compatible Sizes**: 2230, 2242, 2260, 2280
- **Power Supply**: 3.3V, up to 3A (maximum)
- **Indicator Lights**: Includes both power and x4 SSD status LEDs.

1.4 Introduction to Freenove Case Power Board for Raspberry Pi

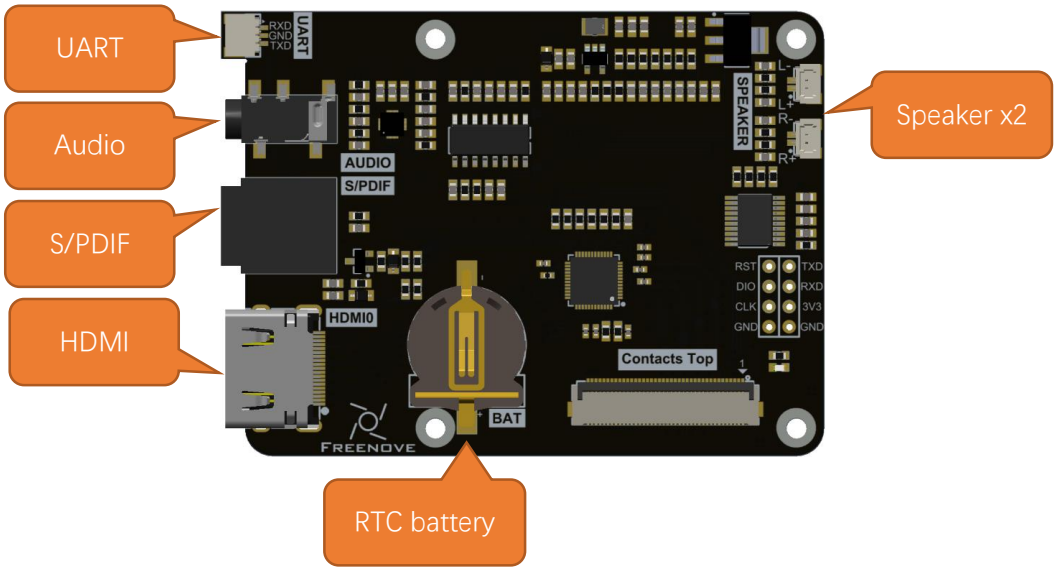
In this tutorial, we name this model as **Audio-Video Board**.



Its core function is to relocate the power connector and HDMI ports to the rear of the case, creating a consolidated I/O layout. It also routes the HDMI0 signal to the Audio-Video Board to enable audio expansion for the case.

1.4 Introduction to Freenove Case Adapter for Raspberry Pi

In this tutorial, we name this model as **Audio-Video Board**.

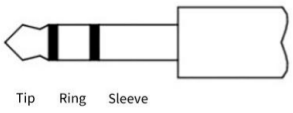
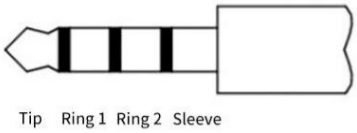
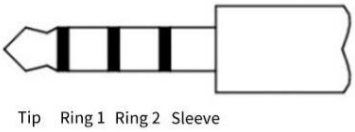


The core function of the Audio-Video Board is to losslessly extract audio from the Raspberry Pi 5's HDMI0 signal, leaving the original video signal intact. The board provides the following interfaces:

- **3.5mm audio jack and two speaker connectors**: Directly play audio from the HDMI source.
- **S/PDIF interface**: For connecting to higher-fidelity audio equipment.
- **RTC battery holder and UART header**: Enhance system clock accuracy and enable debugging/expansion.

3.5mm JACK Audio Interface

In the RPi 5 design, the traditional 3.5mm audio output interface has been removed, and audio signals are now solely output through HDMI. To address this, we have developed an audio separation circuit on the Case Adapter Board. This circuit enables audio to be output via the 3.5mm audio interface on the board, You can use the external 3.5mm TRS headphone jack on the Raspberry Pi 5 to listen to audio output. For proper functionality, ensure your headphone connector is compatible. Below is an explanation of common jack types:

	TRS JACK 	**Tip**	**Left Channel**
		Ring	**Right Channel**
		Sleeve	**GND**
OMTP	TRRS JACK 	**Tip**	**Left Channel**
		Ring1	**Right Channel**
		Ring2	**MIC**
		Sleeve	**GND**
CTIA	TRRS JACK 	**Tip**	**Left Channel**
		Ring1	**Right Channel**
		Ring2	**GND**
		Sleeve	**MIC**

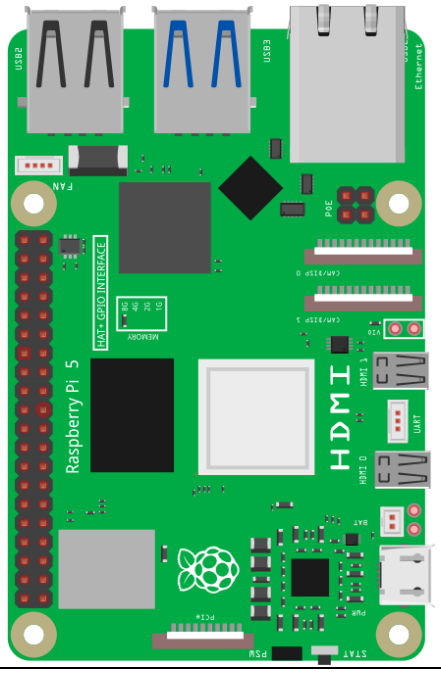
RTC Interface

To enhance the accuracy of the Raspberry Pi's clock, it is advisable to add a battery to the RTC. For this purpose, we have installed a battery holder on the board. If you decide to add a battery, we recommend either purchasing a rechargeable lithium-manganese button-cell battery (with dimensions of 1220 and a voltage of 3V) or an official battery.
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#real-time-clock-rtc

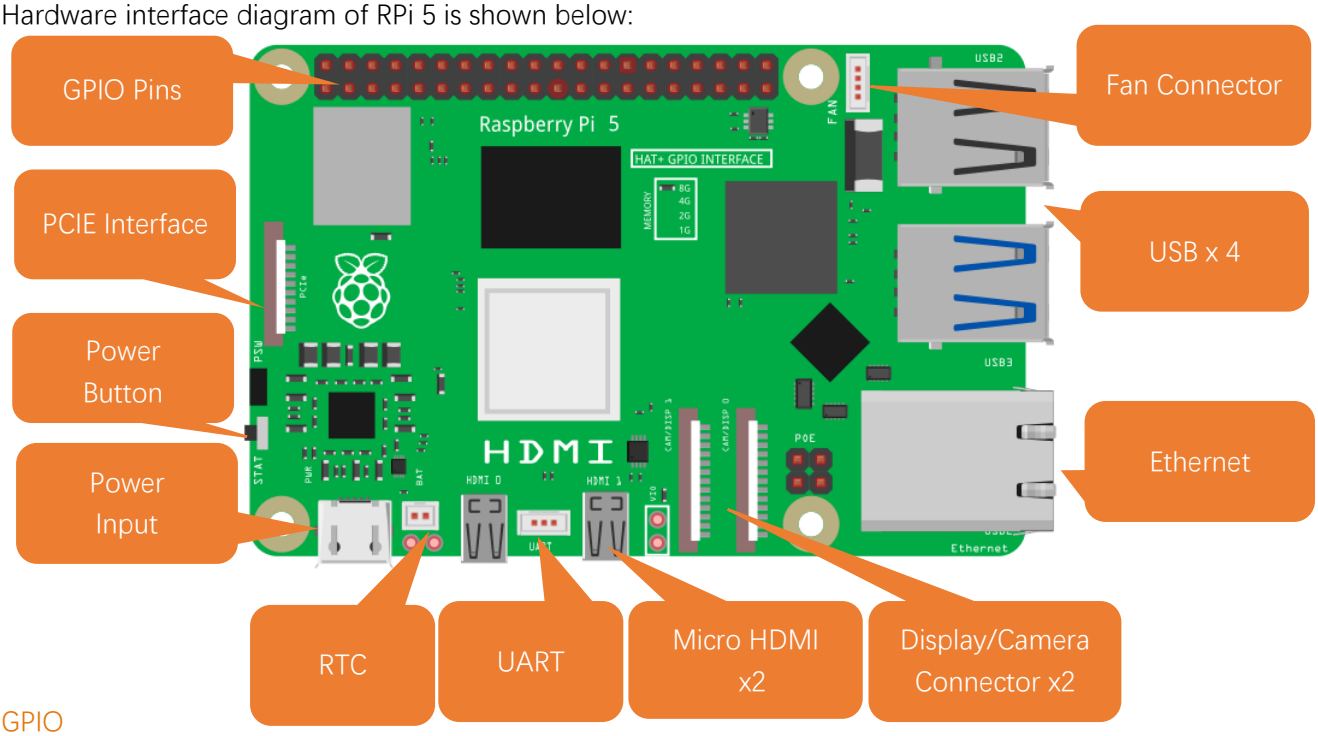
Due to the addition of audio processing capabilities, multiple lighting effects, an enhanced cooling system, and multiple NVMe interfaces (supporting additional SSDs), the overall power consumption during operation has increased. Therefore, only the official 5.1V/5A power adapter (or compatible models with equivalent specifications) should be used to ensure stable system performance.

1.6 Introduction to Raspberry Pi 5 (RPi 5)

At the time of this writing, this product only supports RPi5. The following shows the physical and model figures of an RPi 5.

Practicality picture of Raspberry Pi 5:	Model diagram of Raspberry Pi 5:
	

Hardware interface diagram of RPi 5 is shown below:



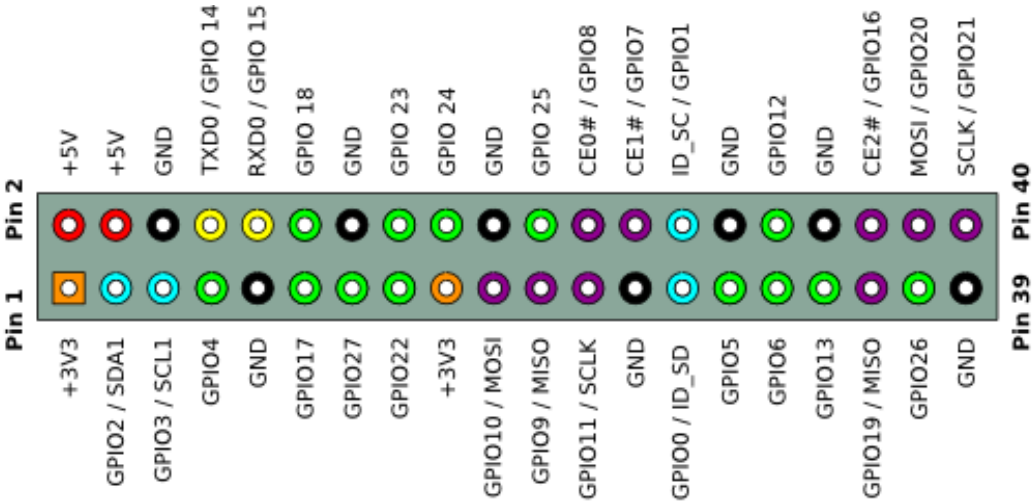
GPIO

GPIO: General purpose input/output. We will introduce the specific feature of the pins on the Raspberry Pi and how you can utilize them in all sorts of ways in your projects. Most RPi Module pins can be used as either an input or output, depending on your program and its functions. When programming the GPIO pins, there are three different ways to reference them: GPIO numbering, physical numbering, WiringPi GPIO Numbering.

BCM GPIO Numbering

The Raspberry Pi CPU uses Broadcom (BCM) processing chips BCM2835, BCM2836 or BCM2837. GPIO pin numbers are assigned by the processing chip manufacturer and are how the computer recognizes each pin. The pin numbers themselves do not make sense or have meaning, as they are only a form of identification. Since their numeric values and physical locations have no specific order, there is no way to remember them, so you will need to have a printed reference or a reference board that fits over the pins.

Each pin is defined as below:



For more details about pin definition of GPIO, please refer to https://pinout.xyz/

Power requirements of various versions of Raspberry Pi are shown in following table:

Product	Recommended PSU current capacity	Maximum total USB peripheral current draw	Typical bare-board active current consumption
Raspberry Pi Model A	700mA	500mA	200mA
Raspberry Pi Model B	1.2A	500mA	500mA
Raspberry Pi Model A+	700mA	500mA	180mA
Raspberry Pi Model B+	1.8A	600mA/1.2A (switchable)	330mA
Raspberry Pi 2 Model B	1.8A	600mA/1.2A (switchable)	350mA
Raspberry Pi 3 Model B	2.5A	1.2A	400mA
Raspberry Pi 3 Model A+	2.5A	Limited by PSU, board, and connector ratings only.	350mA
Raspberry Pi 3 Model B+	2.5A	1.2A	500mA
Raspberry Pi 4 Model B	3.0A	1.2A	600mA
Raspberry Pi 5 Model B	5.0A	1.6A (600mA if using a 3A power supply)	800mA
Raspberry Pi Zero W	1.2A	Limited by PSU, board, and connector ratings only.	150mA
Raspberry Pi Zero	1.2A	Limited by PSU, board, and connector ratings only	100mA

For more details, please refer to
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#power-supply

In this product, the Raspberry Pi 5 is used and it must be powered by a 5.1V/5A power supply. Insufficient power may cause various functions to operate abnormally, or even permanently damage your Raspberry Pi 5. Therefore, we strongly recommend using a 5.1V/5A power supply to ensure optimal performance and avoid potential hardware failure.

Chapter 2 Case Assembly

It is recommended to assemble and use the Freenove Computer Case Kit Mini for Raspberry Pi according to this tutorial. Otherwise, it may lead to incorrect device installation or damage. Please check all the parts again. Before getting started, please check the part list. If any component is missing from your kit, do not start assembly; instead, please email our support team at support@freenove.com to get the missing parts.

Ensure the device is powered off before assembly.

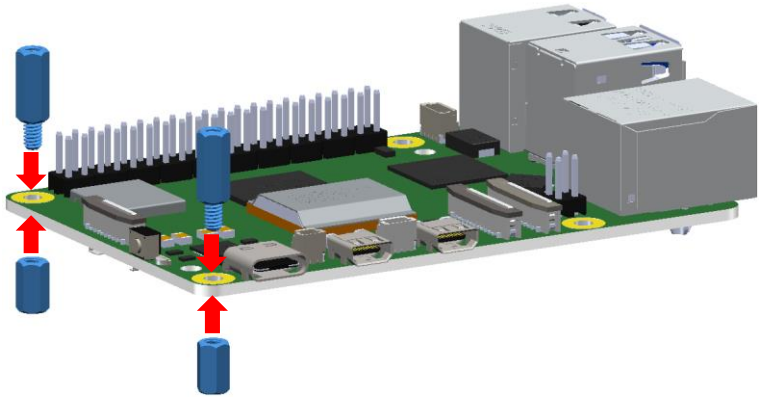
Use of the Screwdriver Bit Holder

When installing the brass standoff, simply place it directly into the bit holder. 
When installing the M3.7x10 Countersunk Head Screws, please use the PH2 hexagonal cross-bit. 
When installing other screws, please use the PH0 hexagonal cross-bit. 

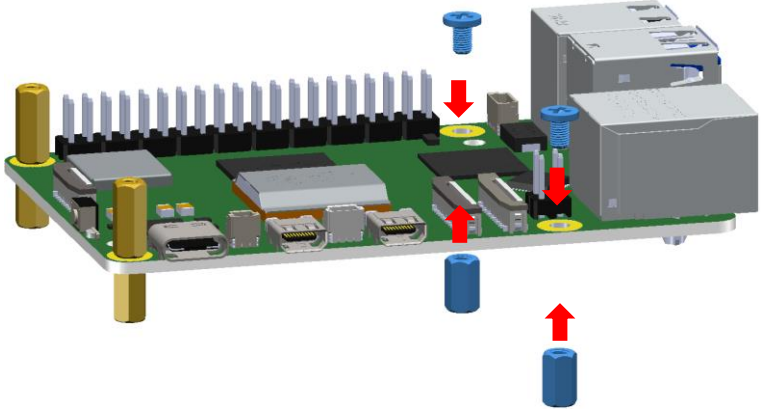
2.1 Raspberry Pi Assembly

2.1.1 Installing Standoffs

Step 1: Install M2.5*10+4 Brass Standoffs and M2.5*7 Brass Standoffs onto the RPi 5.

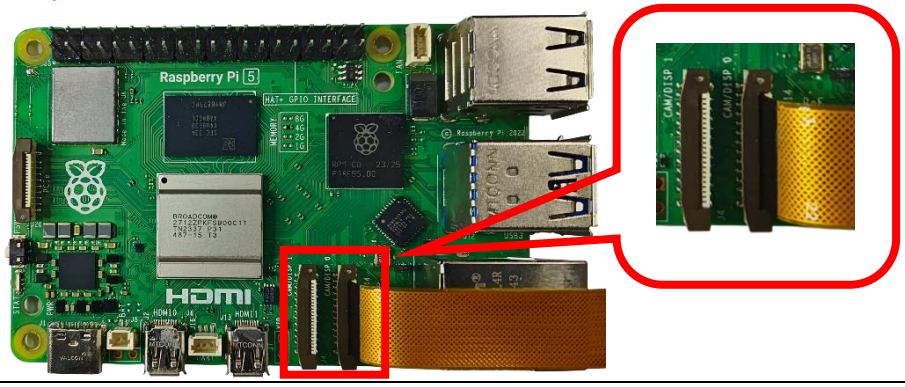


Step 2: Secure the M2.5*7 Brass Standoff onto the RPi 5 with M2.5x4x5 flat-head screws.

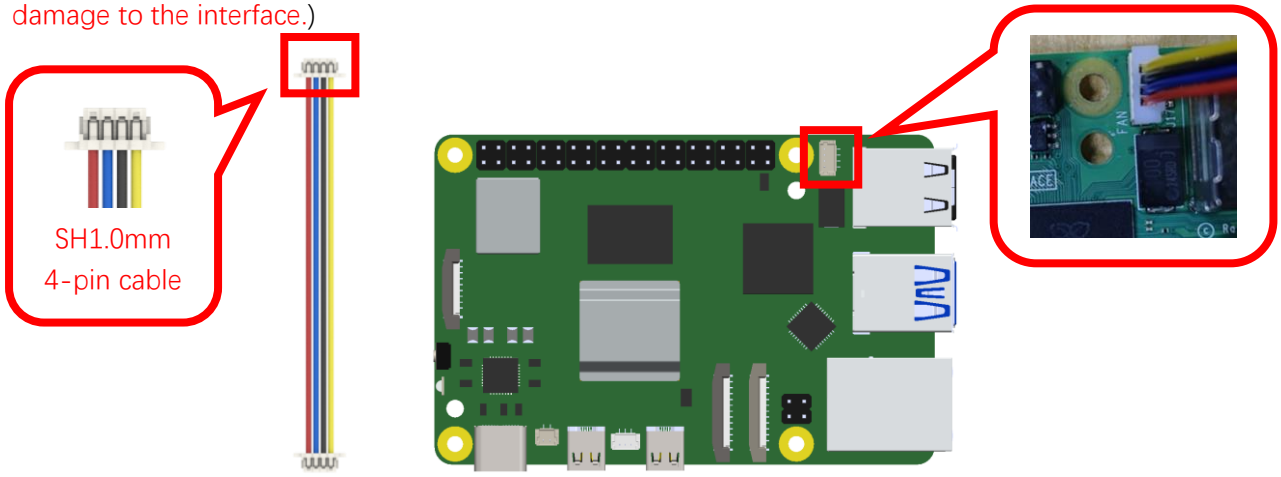


2.1.2 Connecting CAM/DIS Cable and Installing the Tower Cooler

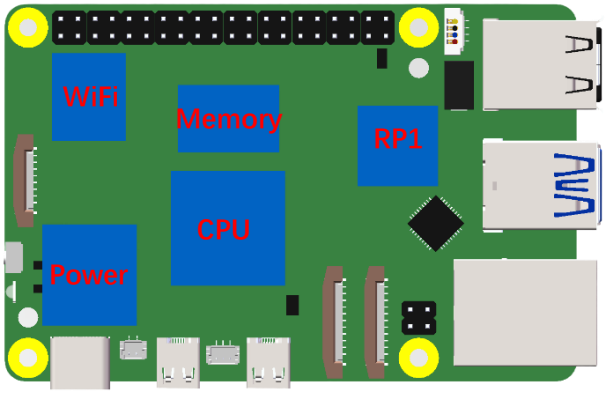
Note: Installing a tower cooler may obstruct access to the CAM/DISP interface. If you need to connect a device (such as a camera) to this interface, you must connect the cable before installing the cooler. Otherwise, you may skip this step.



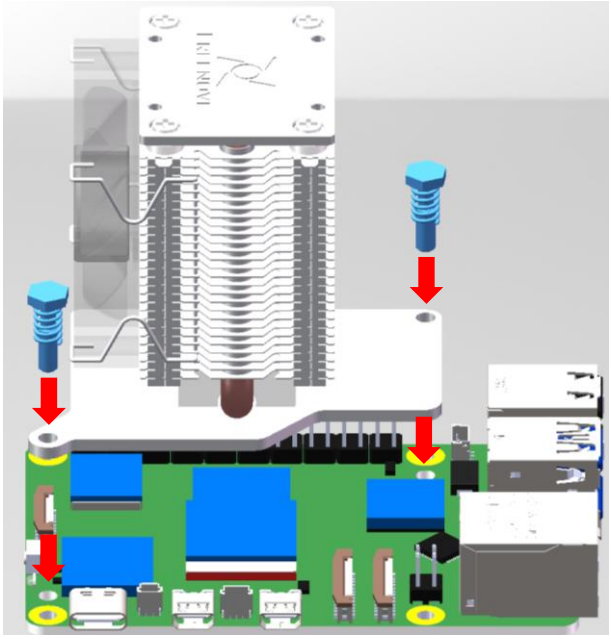
Step 1: Connect the 5cm SH1.0mm 4-pin cable to the FAN (J17) interface on the RPi 5. (This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.)



Step 2: Identify the five thermal pads by size and apply each one to its corresponding chip on the RPi 5. (Important: Ensure you remove the protective film from both sides of each pad to guarantee optimal thermal conductivity.)



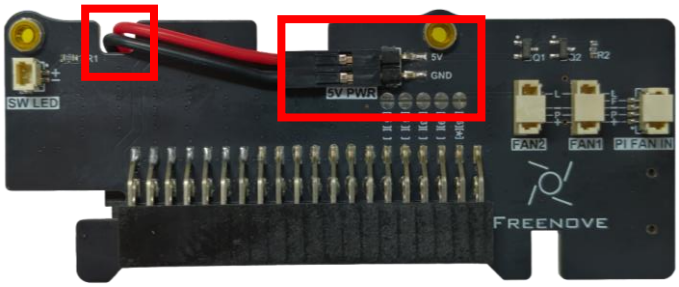
Steps 3: Fix the Tower Cooler to RPi 5 with the Nylon standoffs.



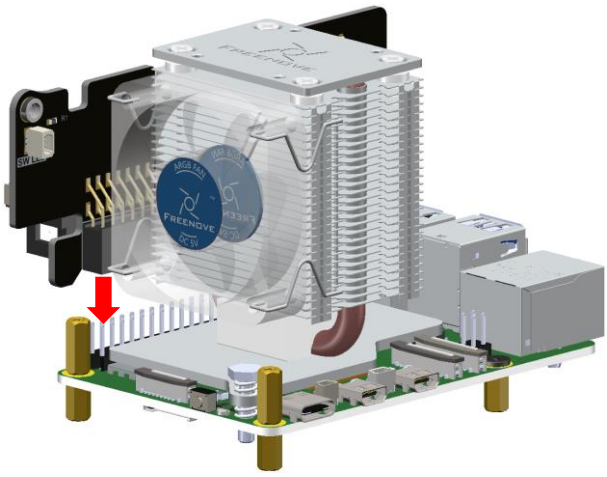
2.2 Mounting GPIO/Switch/NVMe/Power Board

2.2.1 Assembling GPIO Board and Switch Board

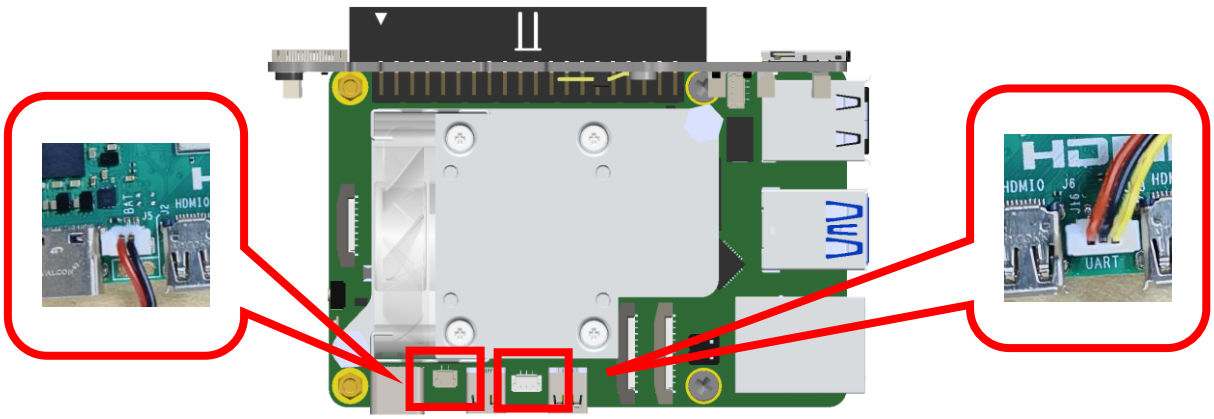
Note: If you plan to use the 5V PWR connector on the GPIO Board to provide additional power to the NVMe Adapter, connect the jumper wires before installing the GPIO Board. A routing notch is provided on the board for this purpose.



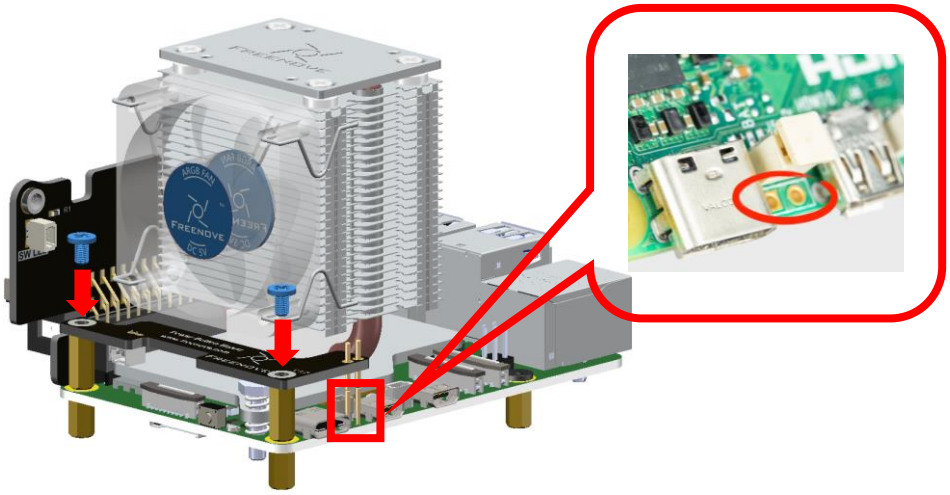
Step 1: Before installing the GPIO Board, ensure that its connector is properly aligned with the Raspberry Pi 5's GPIO header. Once aligned, press it down evenly and firmly. (Note: Correct pin alignment is essential. Misalignment can cause a short circuit and damage the Raspberry Pi.)



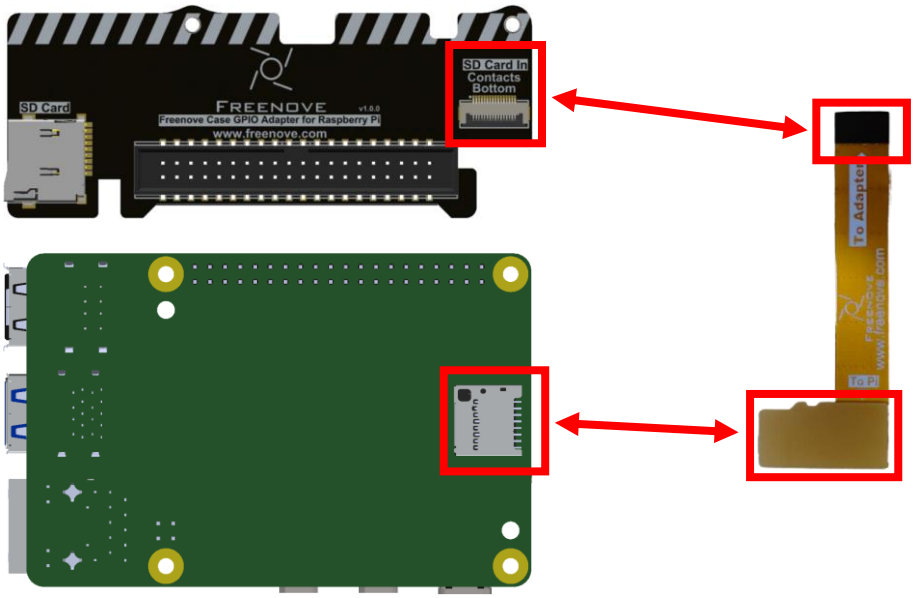
Step 2: Connect the SH1.0mm_2P Same-Direction Cable (5cm) and the SH1.0mm_3P Same-Direction Cable (5cm) to the RTC and UART headers on the Raspberry Pi 5 before installing the Switch Board, **as the Switch Board will block access to the RTC header once installed**.



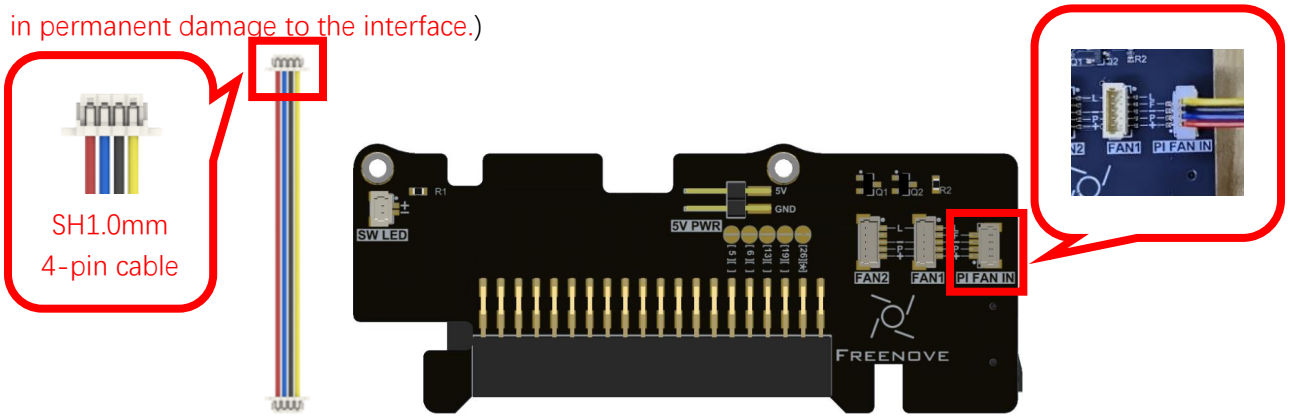
Step 3: Align the pogo pins on the Switch Board with the two vias located between the Raspberry Pi's Type-C and HDMI0 ports. Once properly positioned, secure it using an M2.5x4x5 flat-head screw.



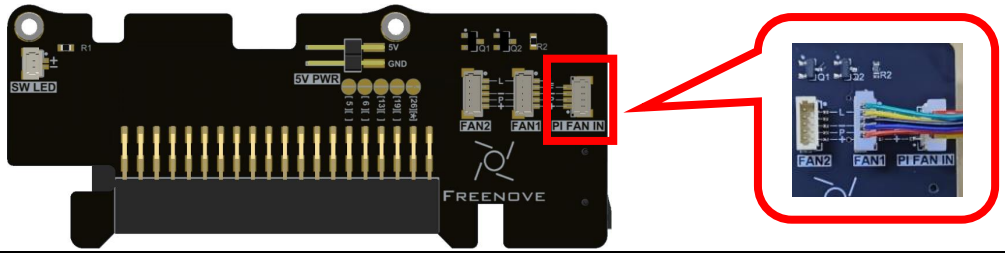
Step 4: Connect SD Card to 0.5mm-16P FPC cable. Insert the "**To Pi**" end into the **SD card slot** on the RPi 5, and insert the "**To Adapter**" end into the "**SD Card In**" interface on the GPIO Board.
(Gently flip-up the clip on the SD card interface to insert the cable.)



Step 5: Connect the other end of the 5cm SH1.0mm 4-pin cable to **the PI FAN IN** header on the GPIO Board. (This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.)



Step 6: Connect the tower coller to the **FAN1** header on the GPIO board. (Note: Speed feedback is only available when a fan is connected to the FAN1 header. This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.)

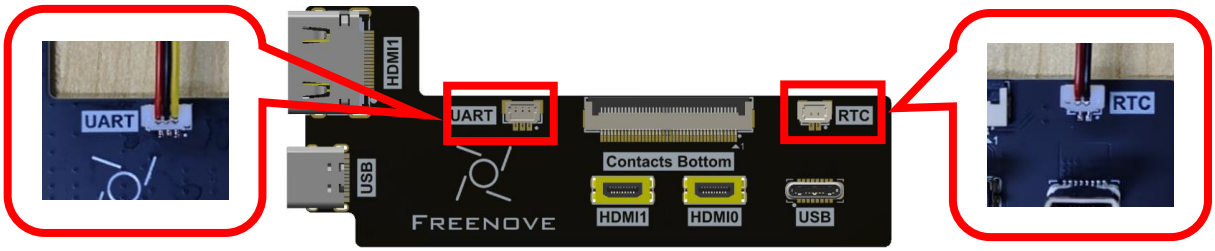


2.2.2 Installing the Power Board

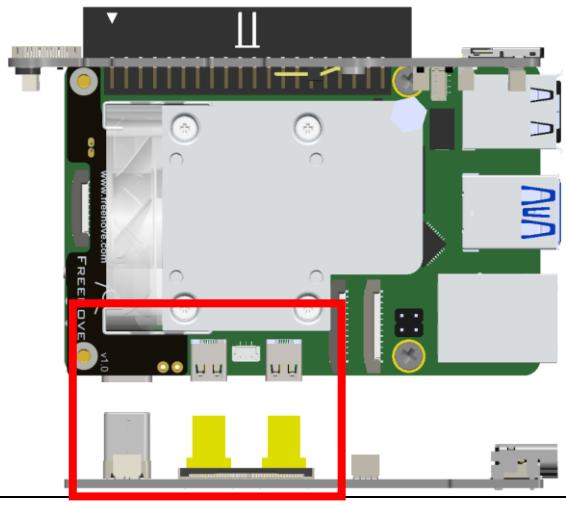
Step 1: Insert one end of the 86mm HDMI FPC Cable into the FPC interface on thePower Board.



Step 2: Connect the other ends of the SH1.0mm_2P Same-Direction Cable (5cm) and the SH1.0mm_3P Same-Direction Cable (5cm) to the **RTC and UART headers** on the Power Board, respectively.



Step 3: Align the Power Board with the Raspberry Pi 5's Type-C, HDMI0, and HDMI1 ports, then insert the Power Board.



2.2.3 Assembling the NVMe Adapter Board

Please select your product model below to view the corresponding installation instructions for the NVMe Adapter Board.

For **FNK0108P/U**: NVMe Adapter Board

For **FNK0108Q/V**: Dual-NVMe Adapter Board

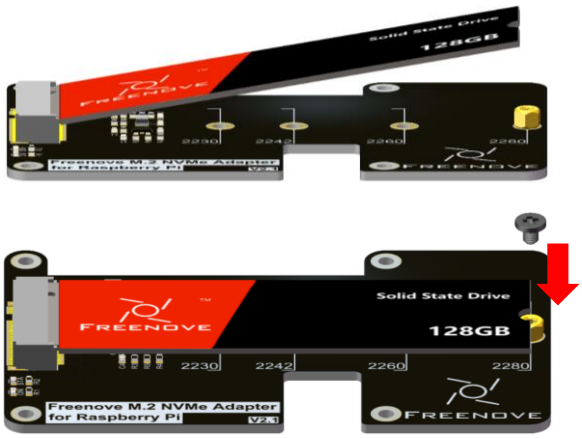
For **FNK0108R/W**: Quad-NVMe Adapter Board

Installing NVMe Adapter Board

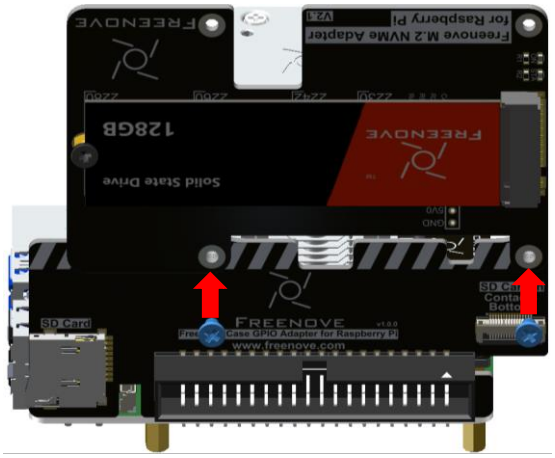
Step 1: Based on the physical dimensions of your SSD, install the M2.5x3+3 brass standoff into the mounting hole corresponding to the 2230/2242/2260/2280 specification. Secure it from the reverse side using an M2.5 nut.



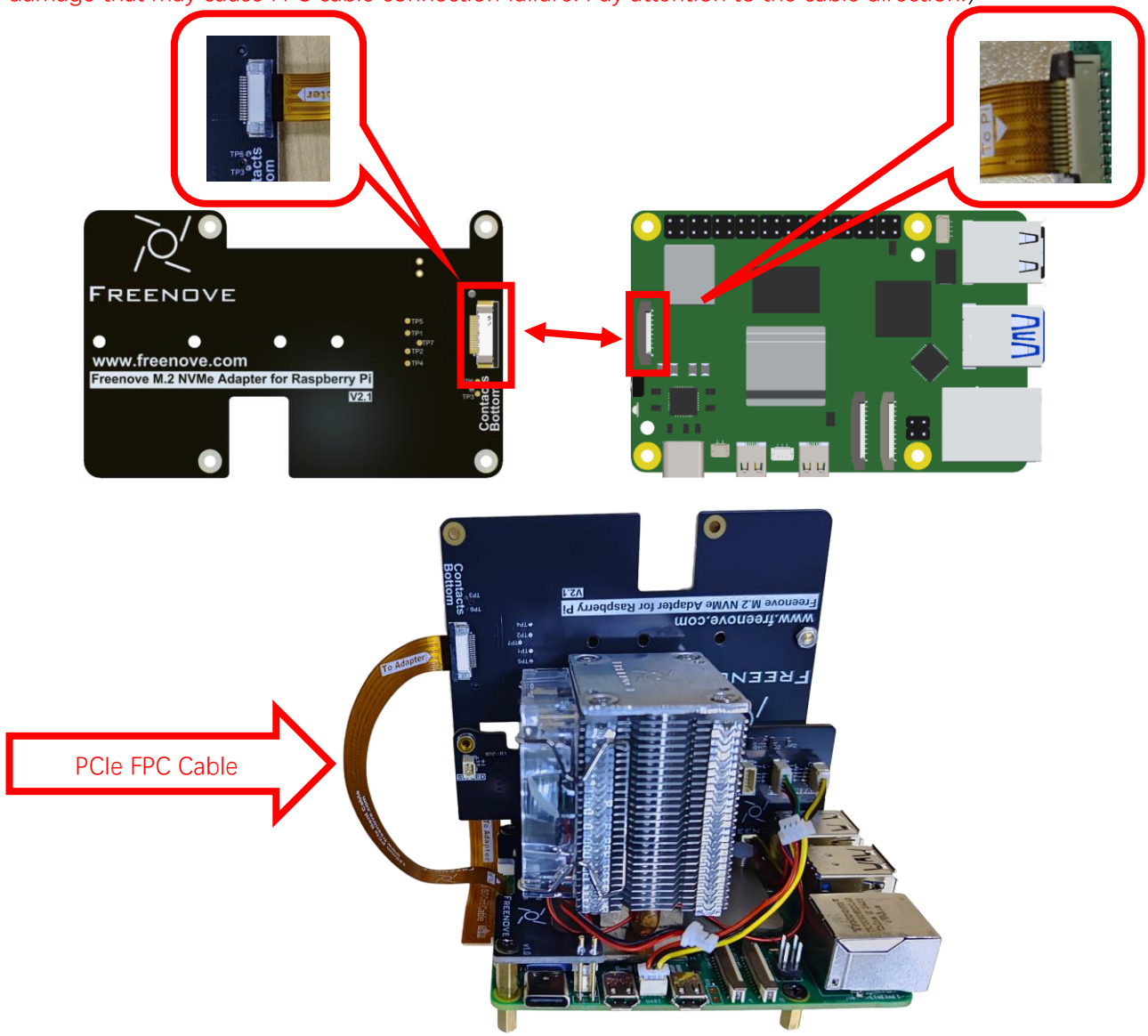
Step 2: Tilt the SSD to insert it into the NVMe slot, and then secure it using an M2.5x2.5x5 flat-head screw.



Step 3: Align the NVMe Adapter Board with the mounting holes on the GPIO Board and secure it with M2.5x4x5 flat-head screws.



Step 4: Connect the PCIe FPC cable. Connect the **To Pi** end to the RPi 5's **PCIe interface**, and the **To Adapter** end to the NVMe Adapter hat's **FPC interface**. (Caution: Gently pull the FPC latch to prevent damage that may cause FPC cable connection failure. Pay attention to the cable direction.)



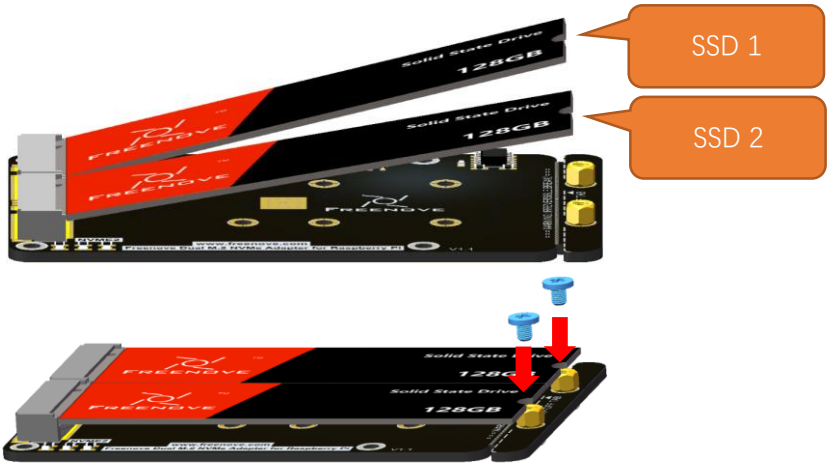
Installing Dual-NVMe Adapter Board

Step 1: Based on the physical dimensions of your SSD, install two M2.5x3+3 brass standoffs into the mounting hole corresponding to the 2230/2242/2260/2280 specification. Secure them from the reverse side with M2.5 nuts.

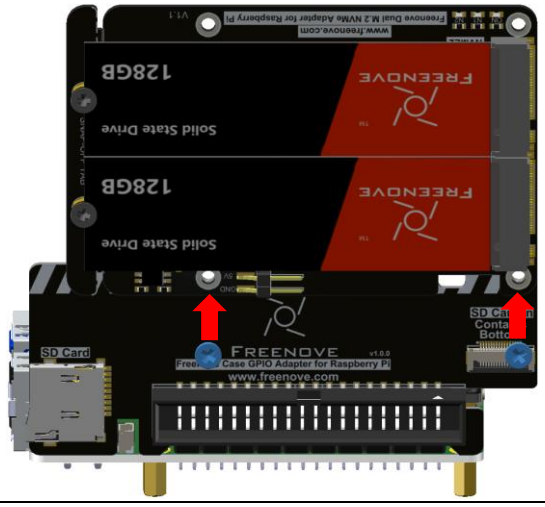


Note: This Adapter Board supports up to two SSDs. Install 0 to 2 SSDs based on your preference.

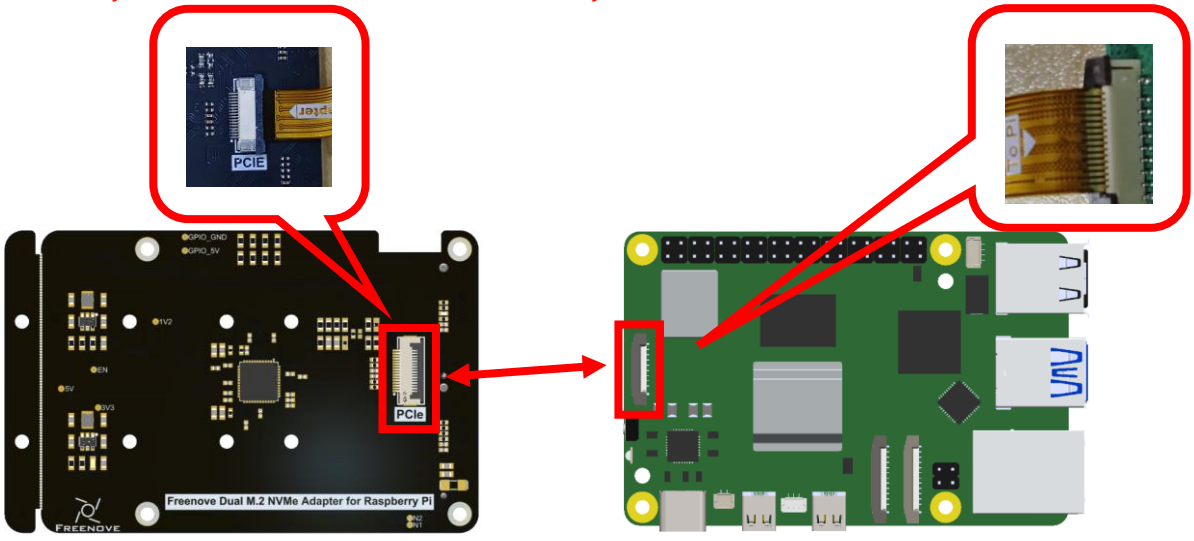
Step 2: Tilt the SSDs to inster them into the two NVMe slots, and then secure them using M2.5x2.5x5 flat-head screws.

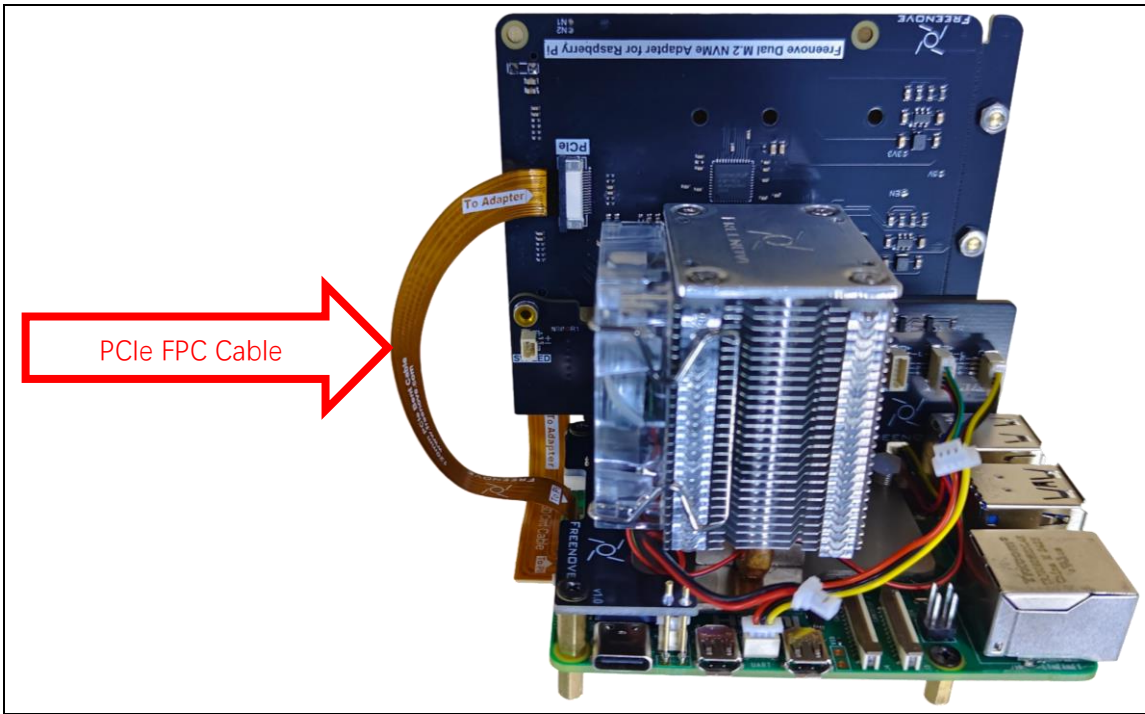


Step 3: Align the Dual-NVMe Adapter Board with the mounting holes on the GPIO Board and secure it with M2.5x4x5 flat-head screws.

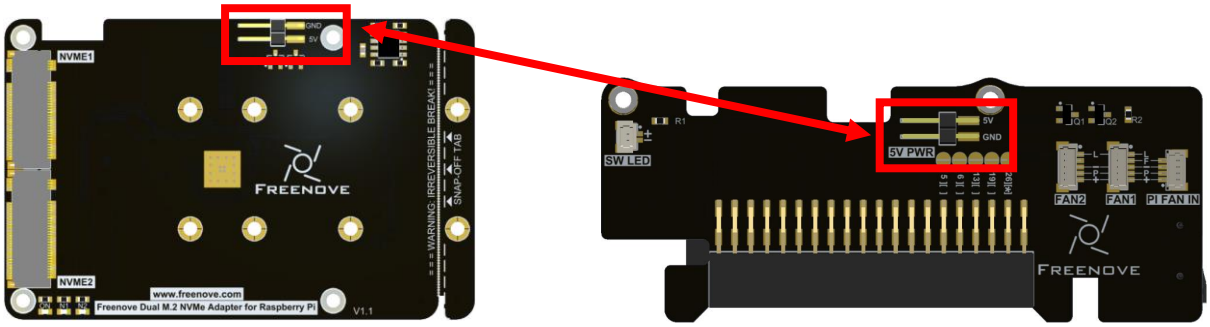


Step 4: Connect the PCIe FPC cable. Connect the **To Pi** end to the RPi 5's **PCIe interface**, and the **To Adapter** end to the NVMe Adapter hat's **FPC interface**. (Caution: Gently pull the FPC latch to prevent damage that may cause FPC cable connection failure. Pay attention to the cable direction.)





Step 5: Connect two female-to-female jumper wires. Attach one end to the **pin headers** on the Dual-NVMe Adapter Board, and the other end to the **5V PWR** on the GPIO Board. (Caution: The red wire must connect to the 5V pin, and the black wire to the GND pin. Incorrect wiring may cause a short circuit and damage the device.)



Generally, connecting the two jumper wires is not required. However, for SSDs with higher power consumption, it is necessary to connect them to ensure adequate power supply and stable operation.

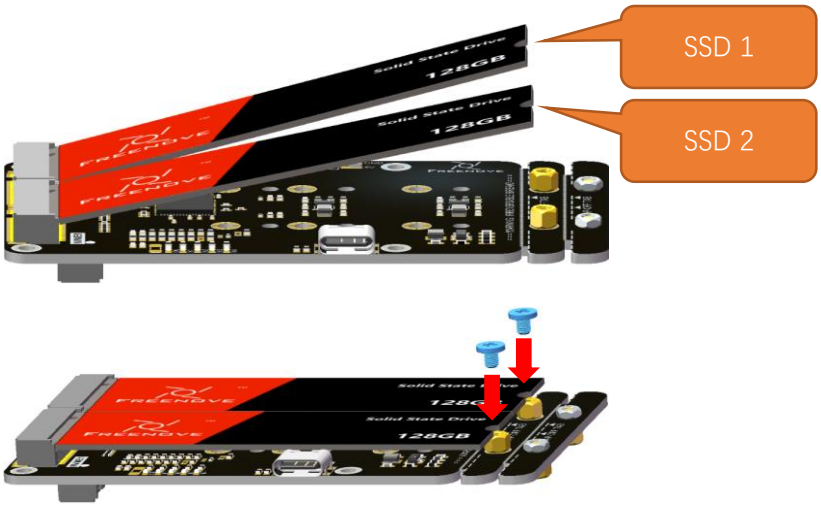
Installing Quad-NVMe Adapter Board

Step 1: Based on the physical dimensions of your SSD, install four M2.5x3+3 brass standoffs into the mounting hole corresponding to the 2230/2242/2260/2280 specification. Secure them from the reverse side with M2.5 nuts. It is recommended that you install standoffs to all the four mounting holes.



Note: This Adapter Board supports up to four SSDs. Install 0 to 4 SSDs based on your preference.

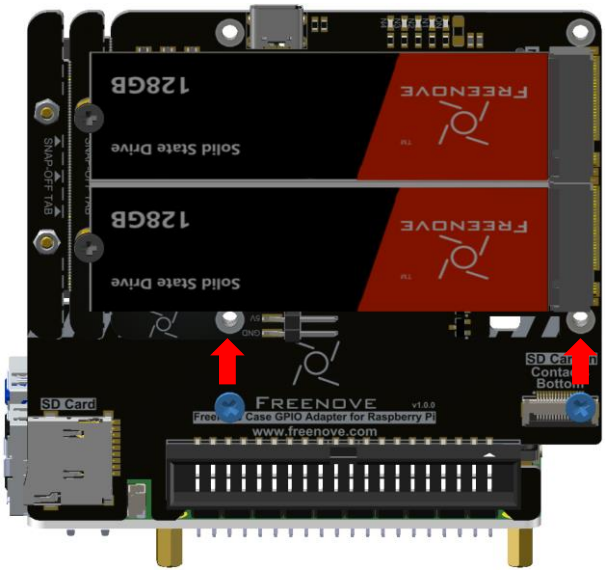
Step 2: Tilt the SSDs to inster them into the two NVMe slots on the front side, and then secure them using M2.5x2.5x5 flat-head screws.



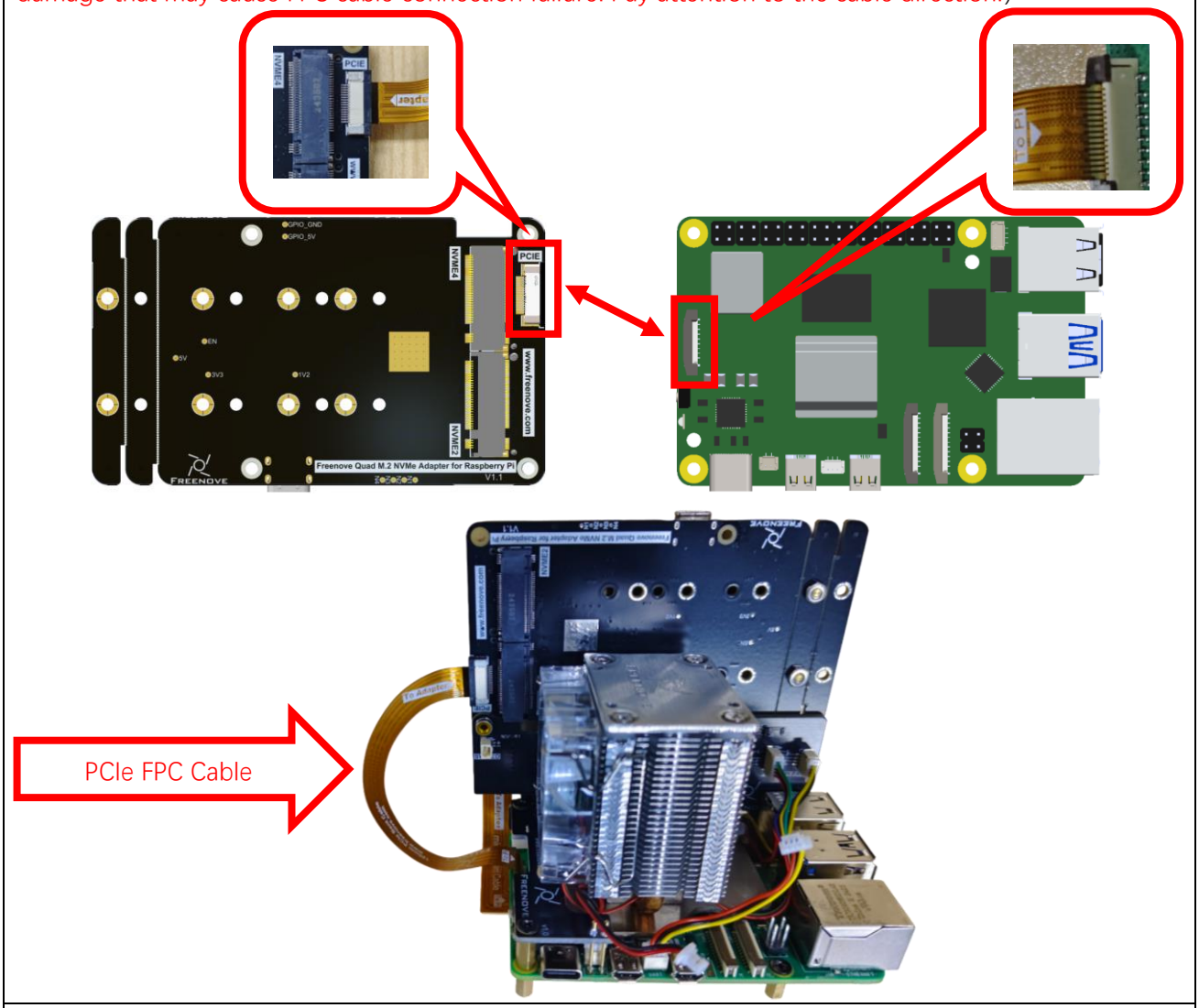
Step 3: Tilt the SSDs to inster them into the two NVMe slots on the back side, and secure them using M2.5x2.5x5 flat-head screws.



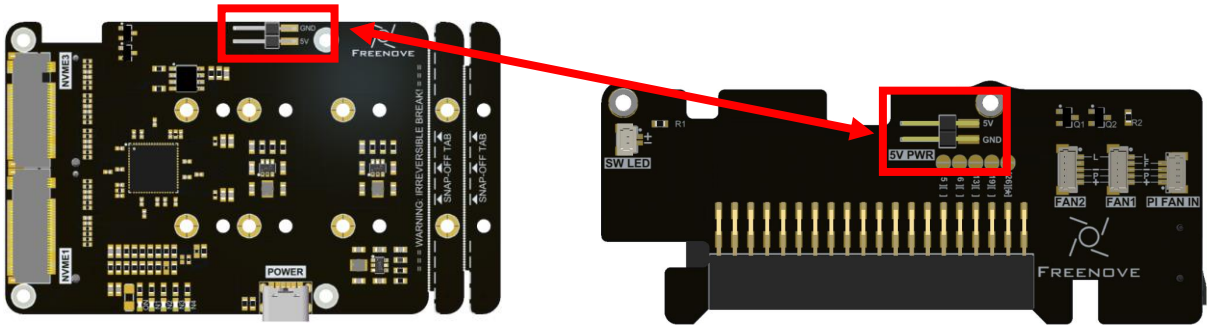
Step 4: Align the Quad-NVMe Adapter Board with the mounting holes on the GPIO Board and secure it with M2.5x4x5 flat-head screws.



Step 5: Connect the PCIe FPC cable. Connect the **To Pi** end to the RPi 5's **PCIe interface**, and the **To Adapter** end to the NVMe Adapter hat's **FPC interface**. (Caution: Gently pull the FPC latch to prevent damage that may cause FPC cable connection failure. Pay attention to the cable direction.)



Step 6: Connect two female-to-female jumper wires. Attach one end to the **pin headers** on the Quad-NVMe Adapter Board, and the other end to the **5V PWR** on the GPIO Board. (Caution: The red wire must connect to the 5V pin, and the black wire to the GND pin. Incorrect wiring may cause a short circuit and damage the device.)



Generally, connecting the two jumper wires is not required. However, for SSDs with higher power consumption, it is necessary to connect them to ensure adequate power supply and stable operation.

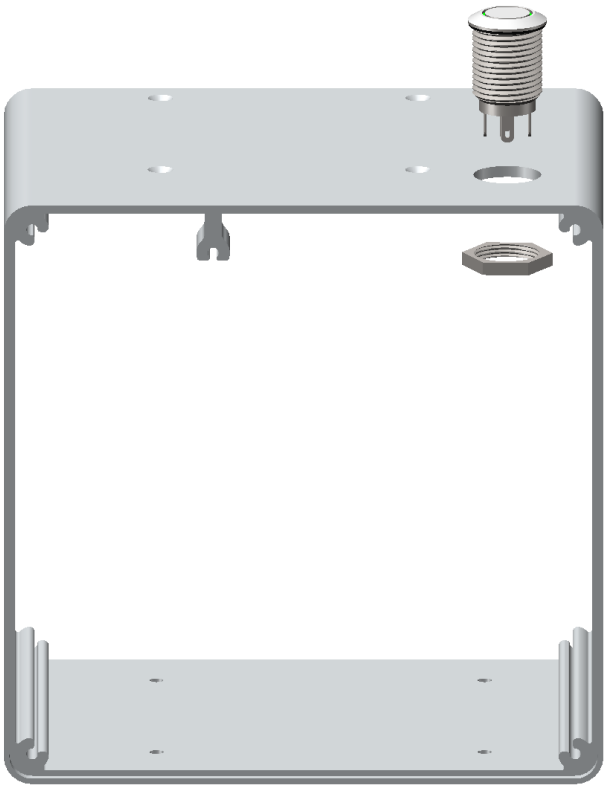
2.3 Case Assembly

2.3.1 Assembling the Power Button and Connecting its cables

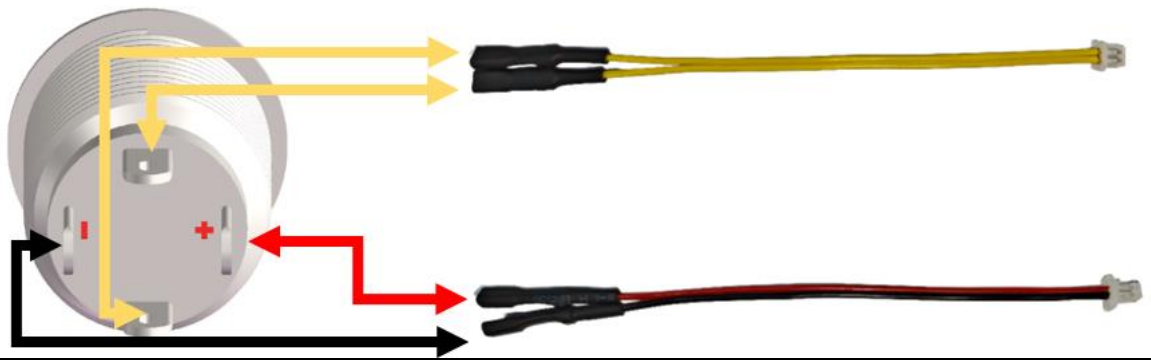
Step 1: Insert the Black Sealing Gasket into the 12mm LED Power Button.



Step 2: Place the assembled 12mm LED Power Button into the circular slot on the top of the case. Secure it from the inside of the case using an M12 nut.

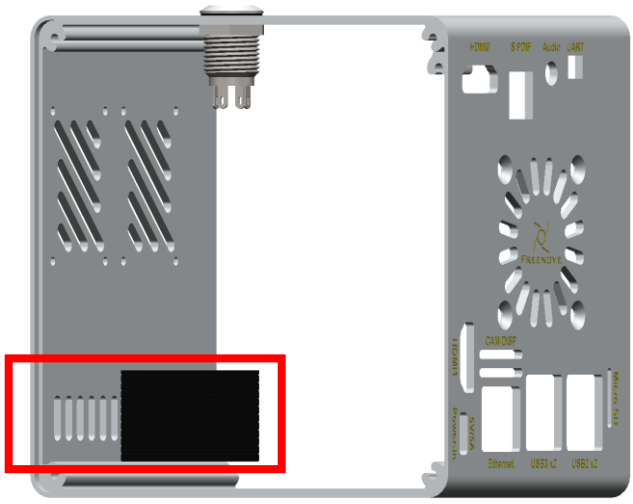


Step 3: Connect the 7mm SH1.0mm to quick-disconnect terminal cable (red/black) and 7mm 1.25mm to 2.8mm quick-disconnect terminal cable (yellow-yellow) to the power headers. (Note: The red wire connects to the switch's '+' terminal, and the black wire connects to the '-' terminal.)



2.3.2 Installing Dust Filter, Fans, and Speakers

Step 1: Peel off the protective film from the air inlet dust filter and affix it to the air inlet on the case.

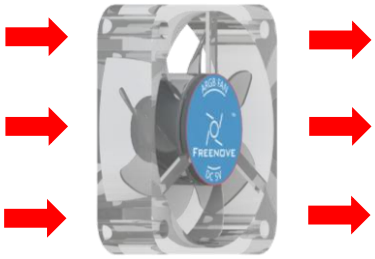


Step 2:

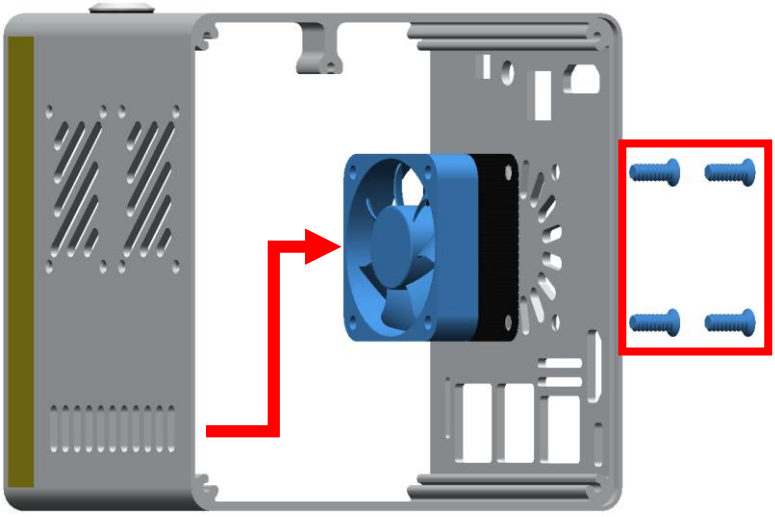
Airflow Direction Description:

The full fan blades are visible, is the **air intake**.

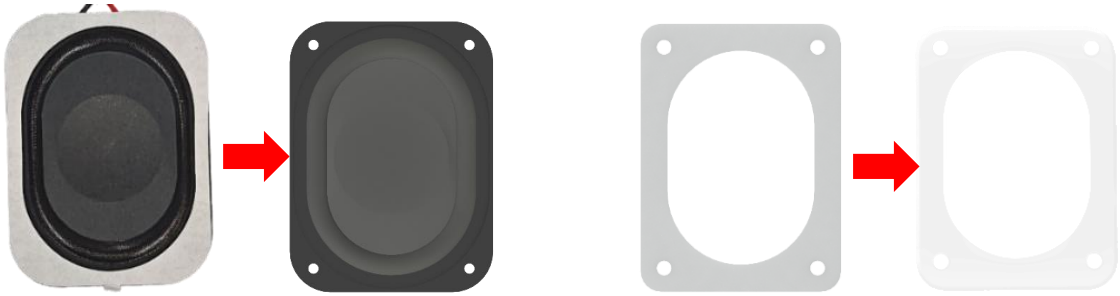
The side with the motor frame and protective grille is the **air outlet**.



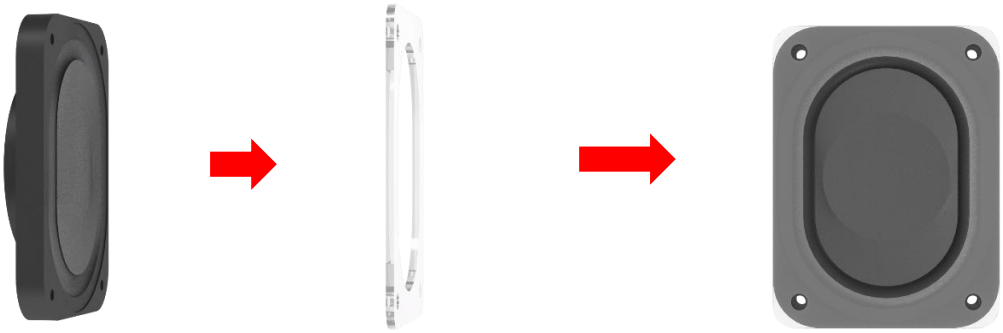
Place the dust filter between the fan and the case. Secure the fan using four M3.7x10 countersunk head screws from the outside of the case. (Note: During installation, ensure the fan's airflow direction is correct, The red arrow indicates the direction of airflow.)



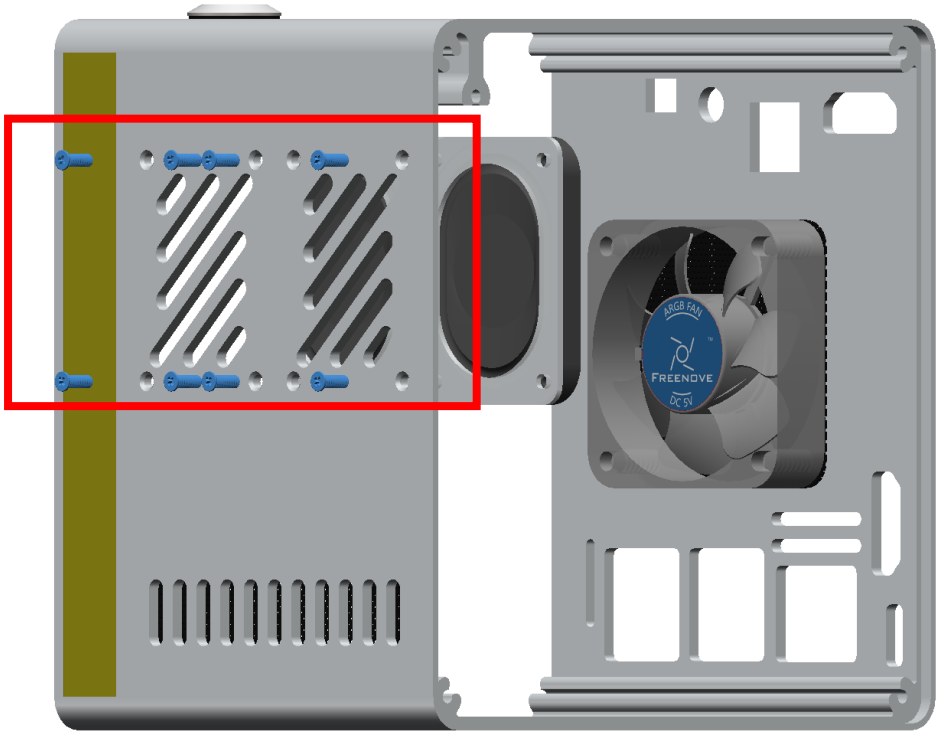
Step 3: Remove the white protective film from the speaker and the speaker's acrylic pad.



Align the speaker acrylic pad with the speaker and attach them together.

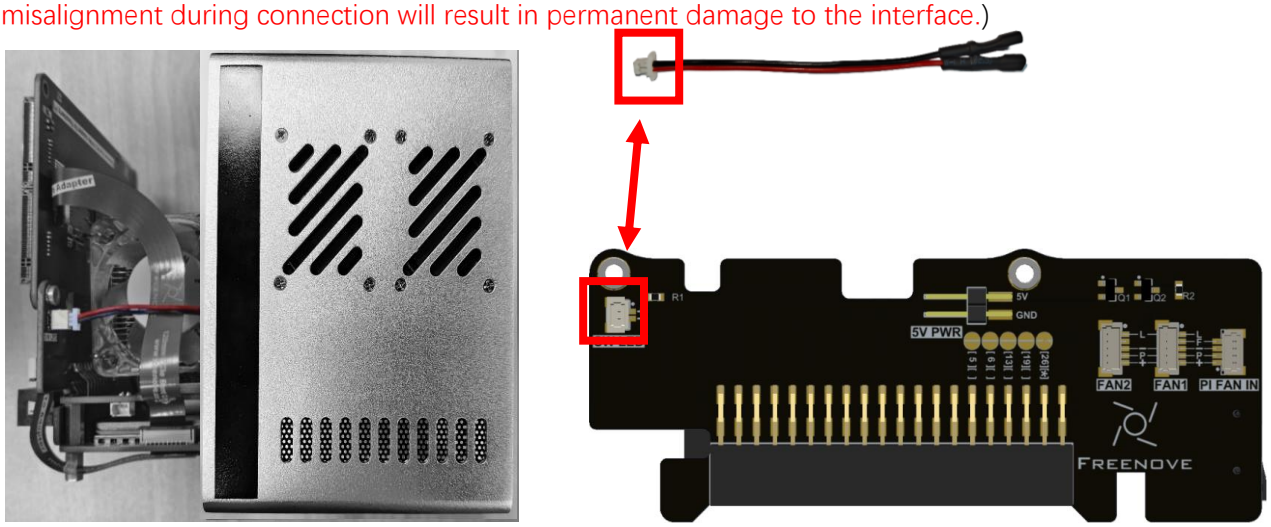


Mount the two speakers to the case frame with M2*6 Countersunk Head Screws.

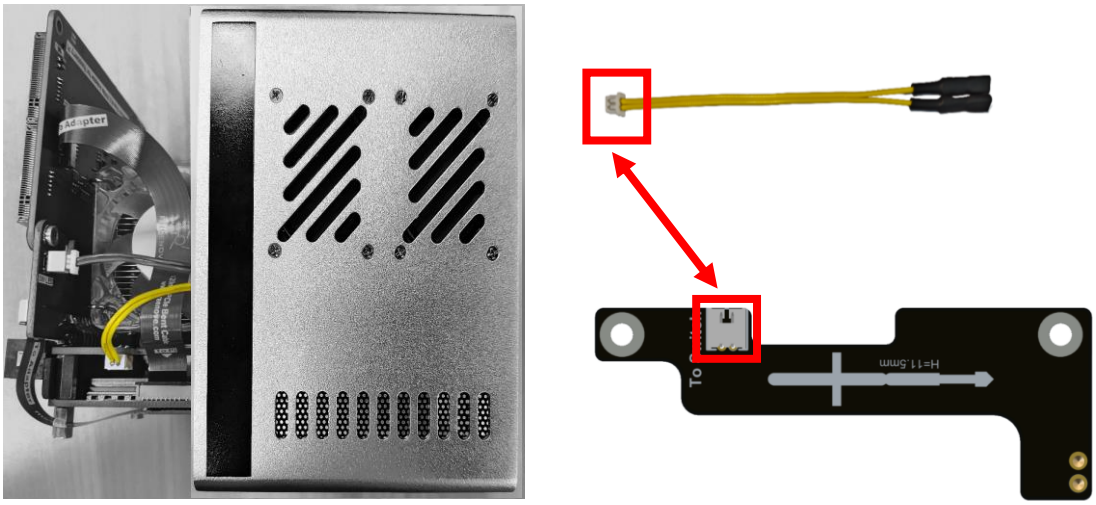


2.3.3 Placing the RPi into the Case

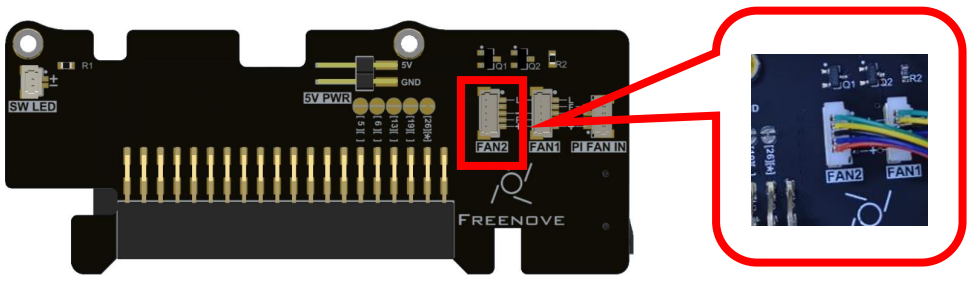
Step 1: Connect the other end of the 7mm SH1.0mm to quick-disconnect terminal cable (red/black)into the **SW LED** header on the GPIO board. (This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.)



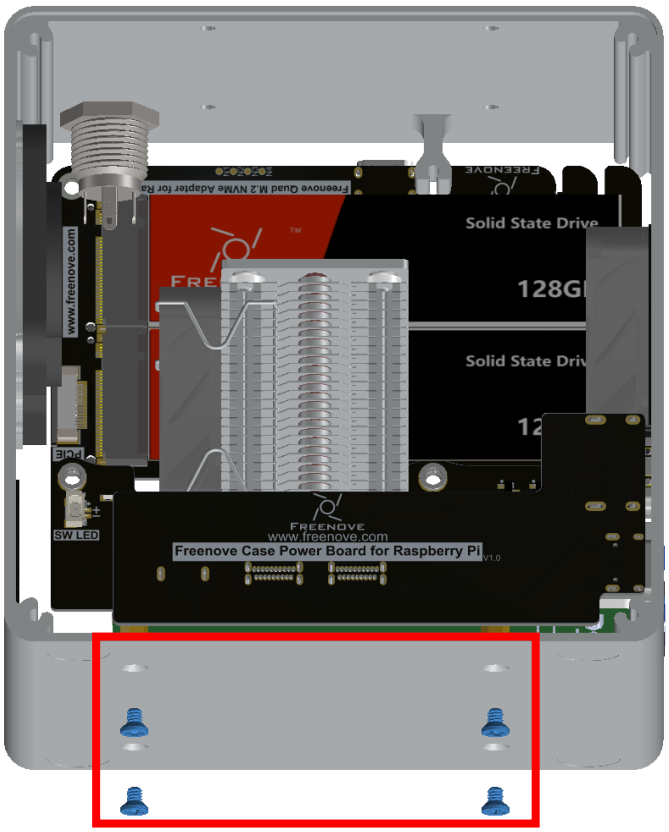
Step 2: Connect the other end of the 7mm 1.25mm to 2.8mm quick-disconnect terminal cable (yellow-yellow) into the Power Board. (This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.)



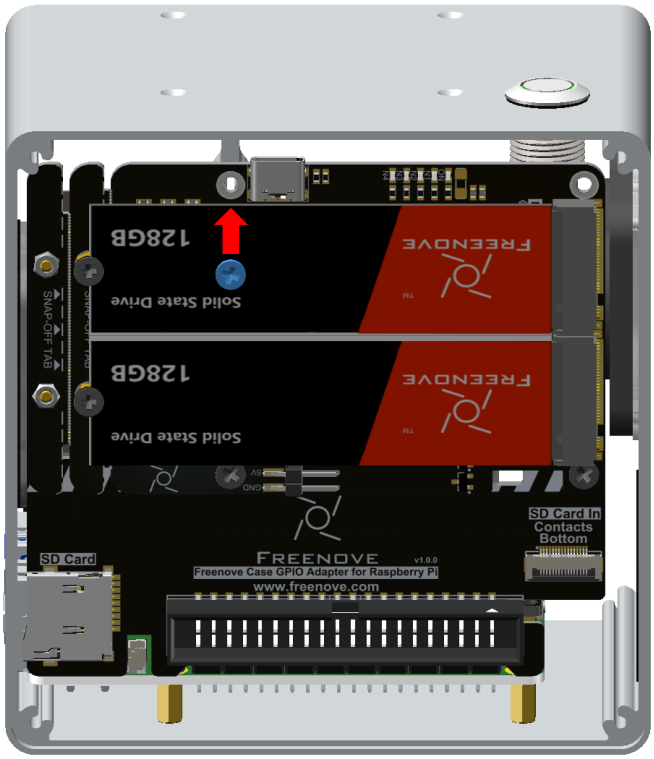
Step 3: Connect the case fan to the GPIO board's **FAN2** header. (This is a non-reversible connector. Improper handling or misalignment during connection will result in permanent damage to the interface.)



Step 4: Slide the Raspberry Pi into the case, align it with the pre-installed standoffs at the bottom, and secure it from the underside of the case with M2.5x4 countersunk head screws.

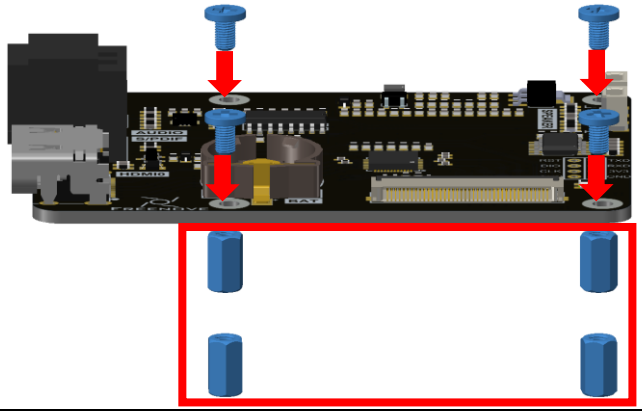


Step 5: Secure the NVMe Adapter board with M2.5x4x5 flat-head screws.

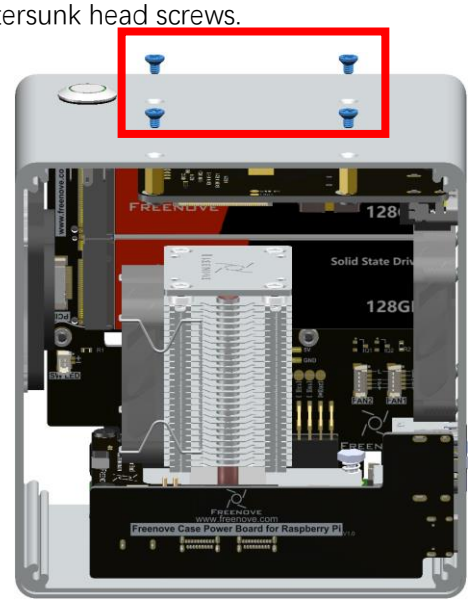


2.3.4 Assembling the Audio-Video Board

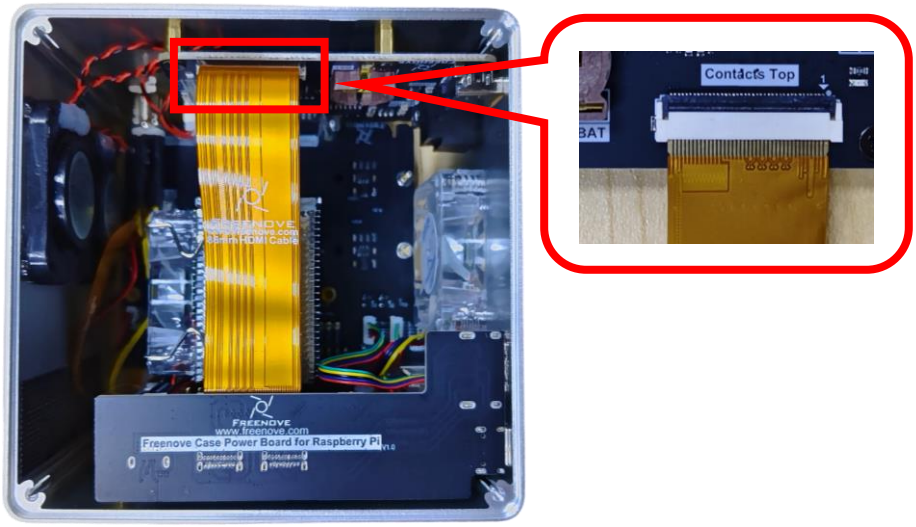
Step 1: Secure the M2.5*7 Brass Standoffs onto the Audio-Video Board with M2.5x4x5 flat-head screws.



Step 2: Align the Audio-Video Board with the pre-installed standoffs at the top of the case, and secure it from the top using M2.5x4 countersunk head screws.

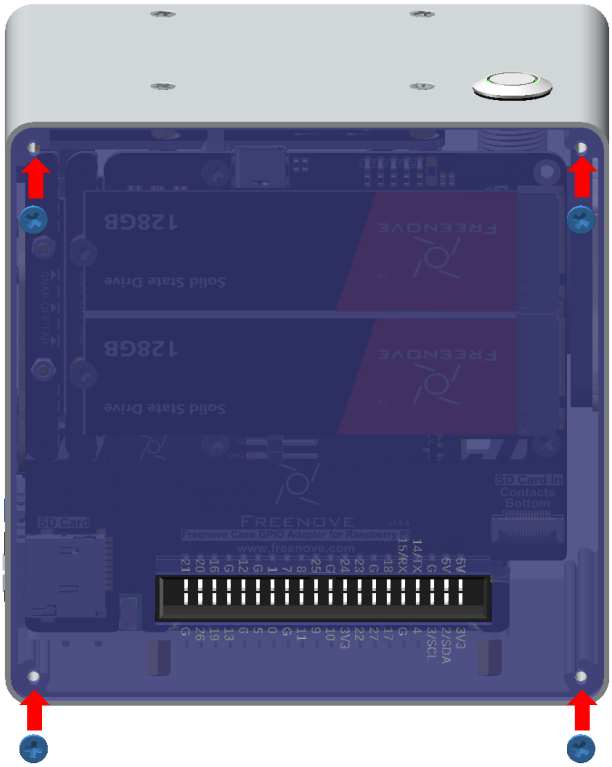


Step 3: Connect the other end of the 86mm HDMI FPC Cable to the Audio-Video Board's FPC interface.

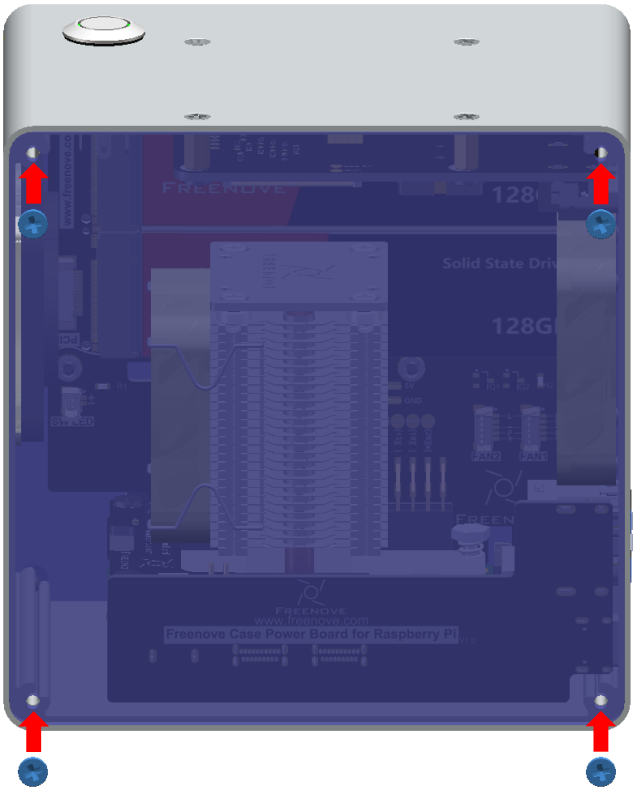


2.3.5 Installing Acrylic Plates and Foot Pads

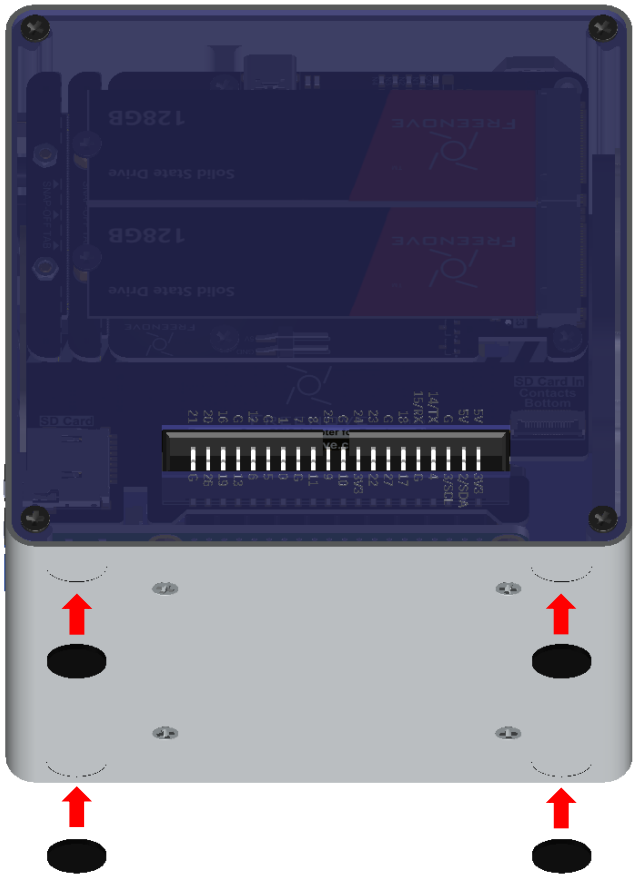
Step 1: Align the acrylic side plate (with the square cutout) with the GPIO headers, and secure it with M2.5x4x5 flat-head screws.



Step 2: Secure the other side plate to the case with M2.5x4x5 flat-head screws



Step 3: Attach the round black non-slip foot pads to the four designated mounting points on the bottom of the case.

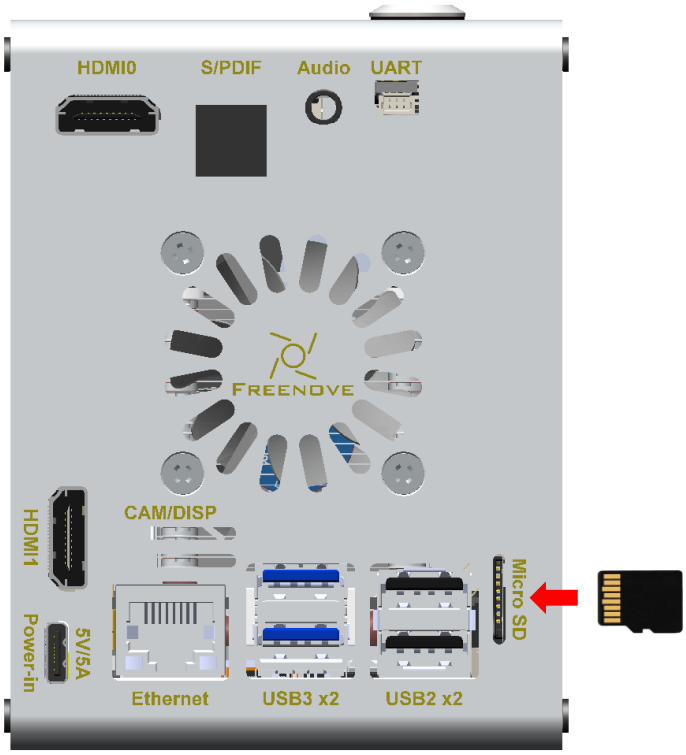


Chapter 3 APP Control

Before powering on the Freenove Computer Case Kit Mini for Raspberry Pi, please make sure that all cable connections are correct.

Due to its multiple functions, this case requires an adequate power supply. We highly recommend using the official Raspberry Pi 5.1V / 5A power adapter (https://www.raspberrypi.com/products/27w-power-supply). Fail to do so may result in the Freenove Computer Case Kit Mini for Raspberry Pi being unusable or causing damage to components.

How to Insert and Take out SD Card



Insert it into the self-ejecting TF card slot on the back of the case with the gold contacts facing up.
Push the SD card in until you hear a click, indicating it is securely locked in place.
To eject the SD card, press it in again until you hear a click, and the card will pop out.

Installing Raspberry Pi OS

If you have not yet installed the Raspberry Pi OS, please refer to the documentation "Installing Raspberry Pi OS.pdf" under the directory **Freenove Computer Case Kit Mini for Raspberry Pi** to install the OS to your SD card or SSD.

Important note for users of the **FNK0108R/Q/V/W** models: due to hardware limitations of the Dual-NVMe and Quad-NVMe Adapter Boards, please **do not** enable **PCIe 3.0** mode, as this may cause the Raspberry Pi to fail to boot.

If you have a display ready, please continue reading the tutorial below.

3.1 Boot Behavior & Environment Settings

3.1.1 What to Expect on First Startup

Upon finishing assembly, please follow the steps below to confirm that all functions are working properly.



1. When the Raspberry Pi starts, the fan should **spin twice** automatically, indicating normal operation.
2. During initial startup, the fan's LED won't light up; it will only illuminate after configuration and execution of the program.

Troubleshooting Guide:

If the fan does not spin upon Raspberry Pi boots up, please Immediately power off the system

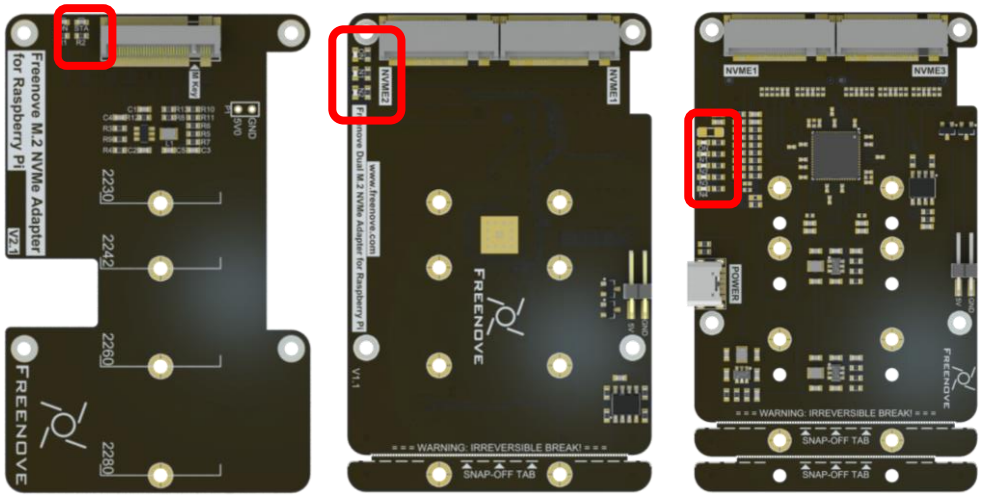
- Verify proper alignment and connection between the GPIO adapter board and Raspberry Pi
- Check all relevant power and data connections

If you have any questions of the above, please contact us at **support@freenove.com**

3. RPi 5 Status LED: The green STAT LED will remain steadily illuminated. If this LED is not lit or displays any pattern other than a steady green light, it indicates that the Raspberry Pi operating system has not booted successfully. In this case, please check your Raspberry Pi hardware and OS installation separately.



4. NVMe Adapter Board Indicators:
   ON LED: Steadily lit.
   STA LED: Blinks in sync with the SSD's built-in activity light.



Should the ON indicator fail to illuminate, please check the cable connection.
Important: It is imperative to utilize the onboard pin header for supplemental power when operating multiple solid-state drives to mitigate risks associated with inadequate power supply.
If all components behave as expected, then your computer case is correctly assembled and functioning well. In this case, you can connect a screen to your Raspberry Pi or access it via VNC viewer.

If you have any questions of the above, please contact us at **support@freenove.com**

3.1.2 Software Setup

Software Packages Update

Run the following command on the terminal to update your Raspberry Pi's package list to the latest version. In the latest Raspberry Pi 13 (Trixie), passwordless sudo is officially disabled. You will need to enter your user password every time you run a sudo command. Please enter your user password when prompted. (Note: The default password is "**raspberry**", please replace it with your user password.)

```
sudo apt update
```

```
pi@raspberrypi:~ $ sudo apt update
[sudo] password for pi:
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://archive.raspberrypi.com/debian trixie InRelease
Hit:4 http://deb.debian.org/debian-security trixie-security InRelease
127 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Code downloading

Open the Raspberry Pi Terminal, type the following two commands one by one to download the code for the case.

```
cd
git clone
https://github.com/Freenove/Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi.git
```

```
pi@raspberrypi:~ $ cd
pi@raspberrypi:~ $ git clone https://github.com/Freenove/Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi.git
Cloning into 'Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi'...
remote: Enumerating objects: 24, done.
remote: Counting objects: 100% (24/24), done.
remote: Compressing objects: 100% (22/22), done.
remote: Total 24 (delta 0), reused 24 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (24/24), 9.09 MiB | 4.00 MiB/s, done.
```

Creating Desktop Shortcut

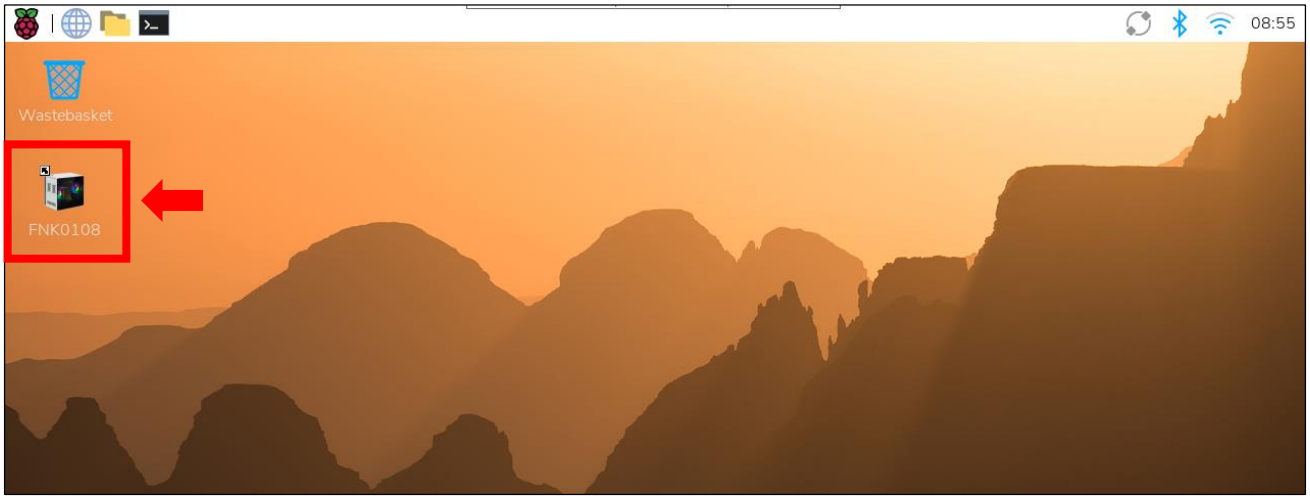
Run the following two commands one by one in the Raspberry Pi terminal to create a desktop shortcut for the case control software. Configuring the software environment takes some time, please wait with patience.

```
cd Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi
sudo python setup.py
```

```
pi@raspberrypi:~ $ cd Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi
pi@raspberrypi:~/Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi $ sudo python setup.py
Starting setup...
Running: sudo apt update
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Get:3 http://archive.raspberrypi.com/debian trixie InRelease [54.9 kB]
Hit:4 http://deb.debian.org/debian-security trixie-security InRelease
Get:5 http://archive.raspberrypi.com/debian trixie/main arm64 Packages [399 kB]
Fetched 454 kB in 3s (169 kB/s)
```

When configuring the environment, you need to select the corresponding product. Please enter "**1**" to download the code for the corresponding product.

```
Please select your kit type:
1. FNK0108: Freenove Computer Case Kit Mini for Raspberry Pi
2. FNK0113:  Freenove Tower Cooler for Raspberry Pi
Enter your choice (1 or 2): 1
```

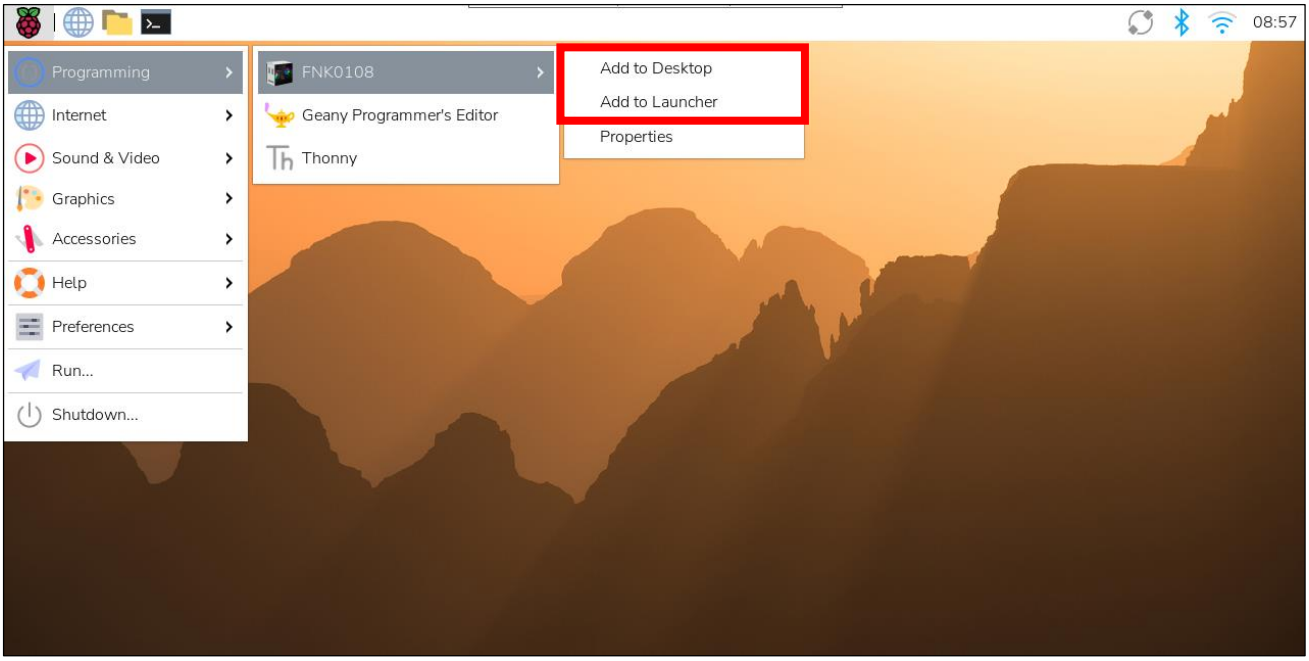


The icon above corresponds to the command shown below.

```
cd Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi/Code/
python app_ui.py
```

```
pi@raspberrypi:~ $ cd Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi/Code/
pi@raspberrypi:~/Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi/Code $ python app_ui.py
```

After completing this step, you will find the application in the "**Programming**" section of the menu. If you accidentally delete the desktop shortcut or wish to add the app to the top launcher, simply right-click "**FNK0108**" and select "**Add to Desktop**"or "**Add to Launcher**".

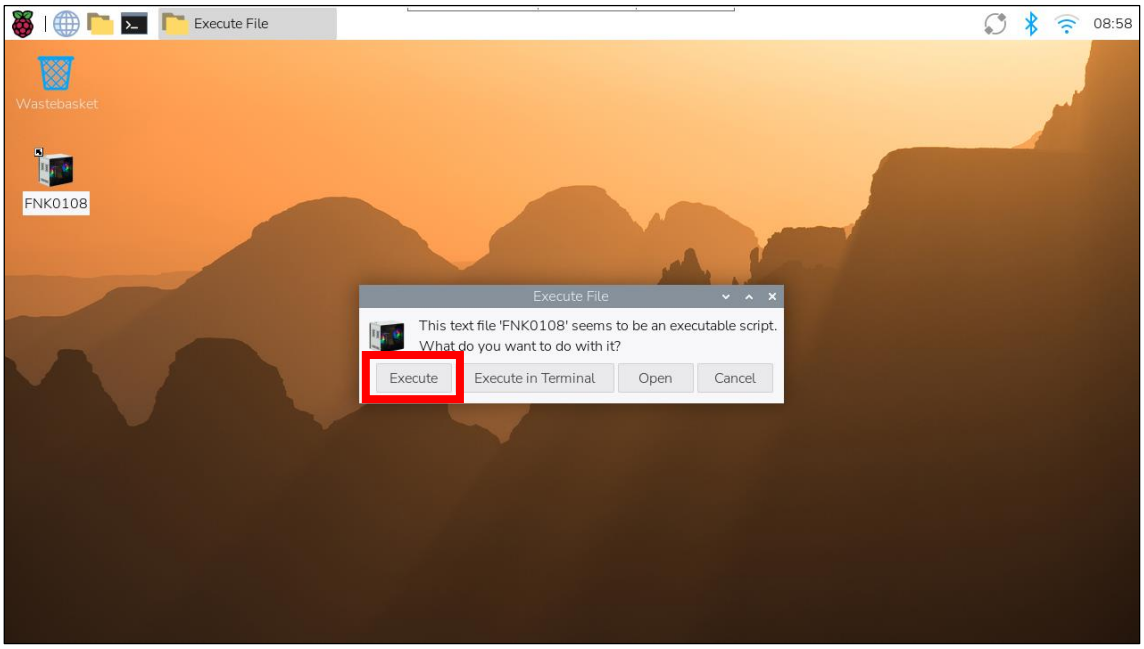


If you are interested in the code implementation, you can explore the files in the Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi/Code directory.
Should you wish to modify the code, please ensure you back it up first to avoid potential software malfunctions caused by unintended changes.

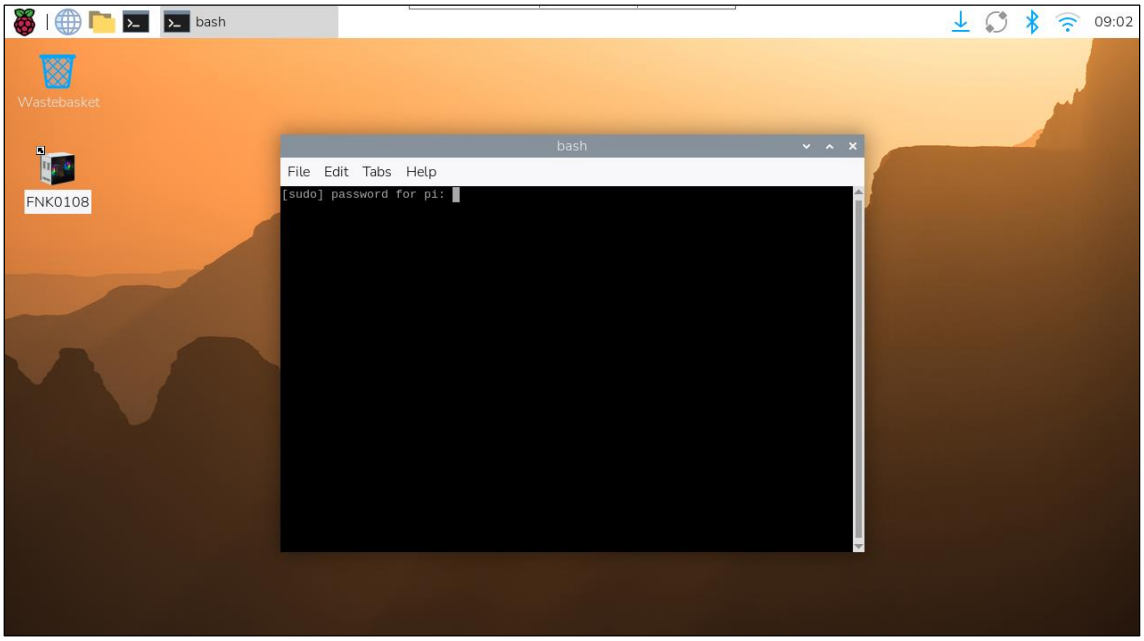
3.2 About the Case Control Software

With the environment configured from the previous chapter, the accompanying host software can now be used to manage case functions including ARGB lighting and fan control.
This chapter provides a detailed guide on the software's usage. For insight into the interface design, the app_ui.py file is available in the Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi/Code directory.

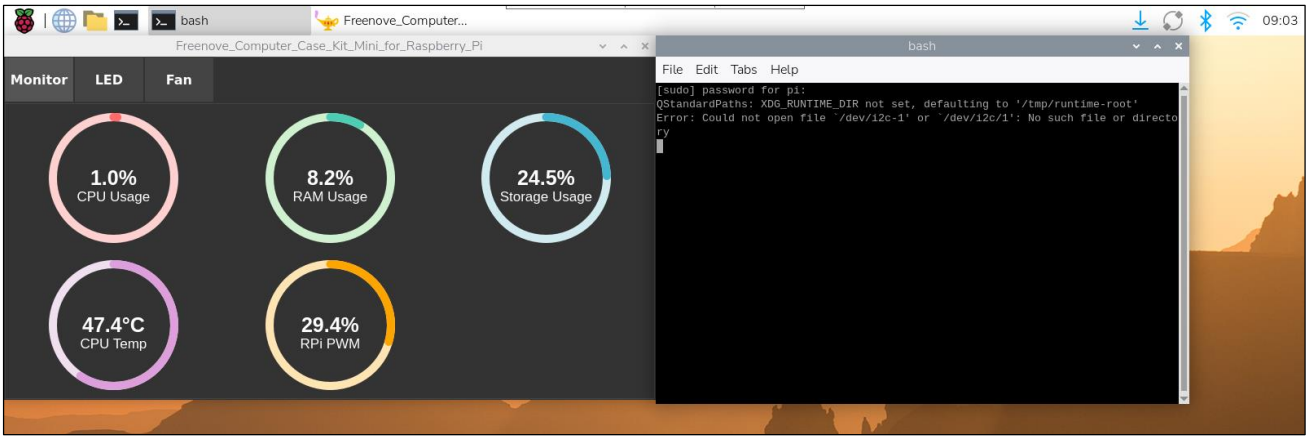
Double click the software with Freenove logo on RPi's desktop, and a window will pop up. Click "**Execute**" to run the program.



Since passwordless sudo is disabled in Trixie, you will also be asked for your password again after reboot. Enter your user password in the pop-up terminal window.

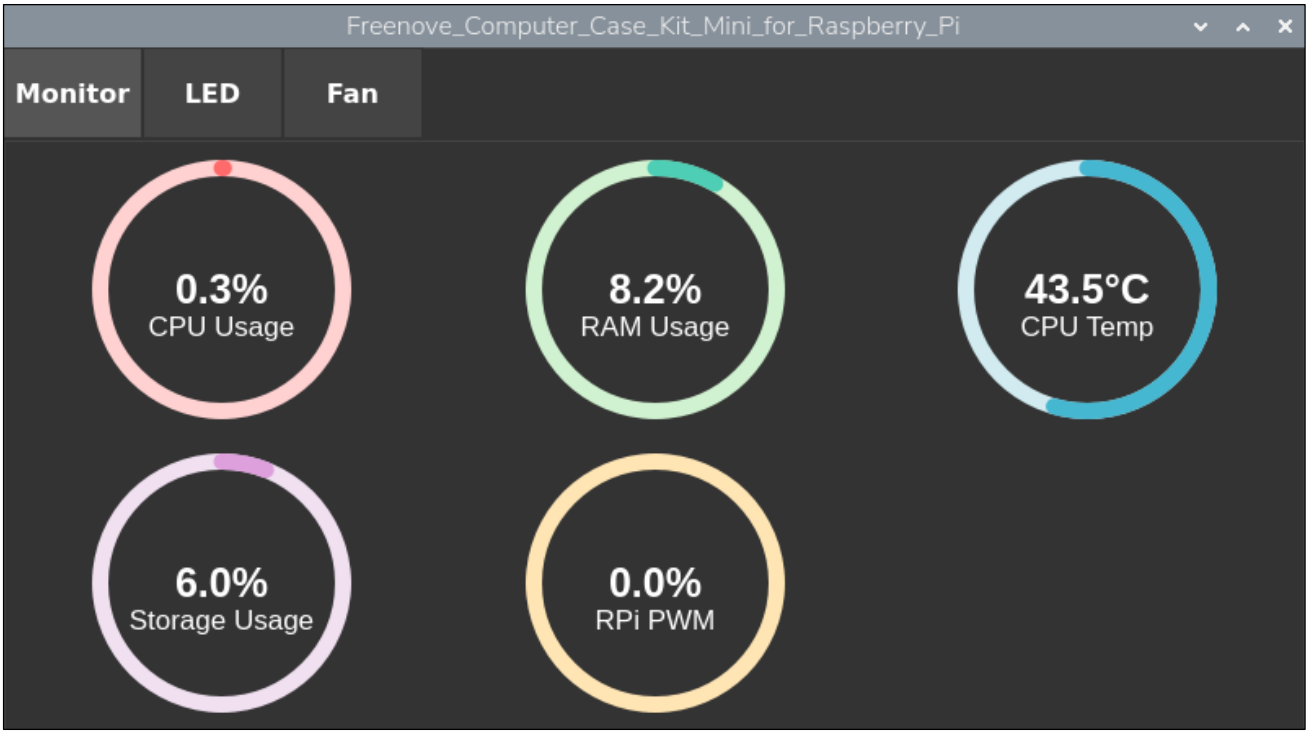


The software interface is as shown below. The left window is the main software interface, and the right window displays debugging information.



3.2.1 Dashboard Monitoring

The dashboard provides live monitoring of key RPi 5 and case component stats, giving you an at-a-glance view of the system status.



Below is a detailed explanation of the parameters displayed on the monitoring interface:

- **CPU Usage**: Current processor utilization percentage of the Raspberry Pi 5
- **RAM Usage**: Current system memory utilization percentage of the Raspberry Pi 5
- **CPU TEMP**: The internal temperature of the Raspberry Pi 5's SoC (System on Chip)
- **Storage Usage**: The total storage space utilization. This value reflects the usage of the SD card if one is used. If single or multiple SSDs are installed, it calculates the aggregate usage across all drives.
- **RPi PWM**: PWM duty cycle for the Raspberry Pi's active cooler fan

Data Source & Troubleshooting:

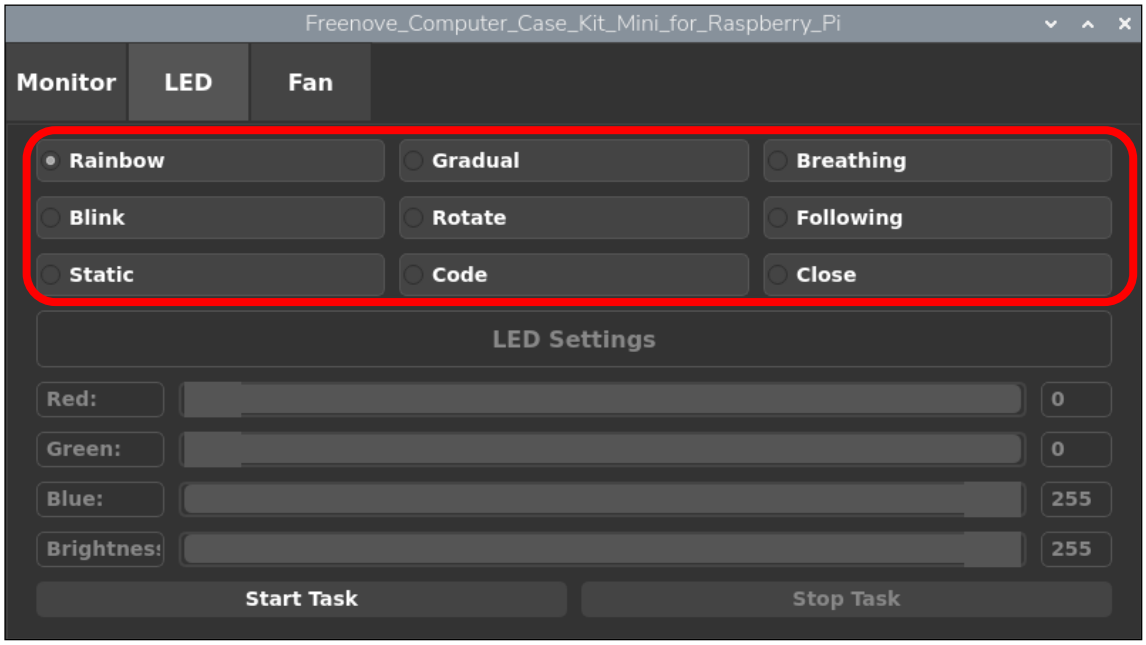
CPU Usage, RAM Usage, CPU Temp, Storage Usage, and RPi PWM are retrieved directly from the Raspberry Pi. If any of these values are missing, check the Raspberry Pi for faults.

If these data cannot be obtained consistently, please contact us by email: **support@freenove.com**

For those interested in the interface implementation, please refer to the files api_systemInfo.py in the Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi/Code directory.

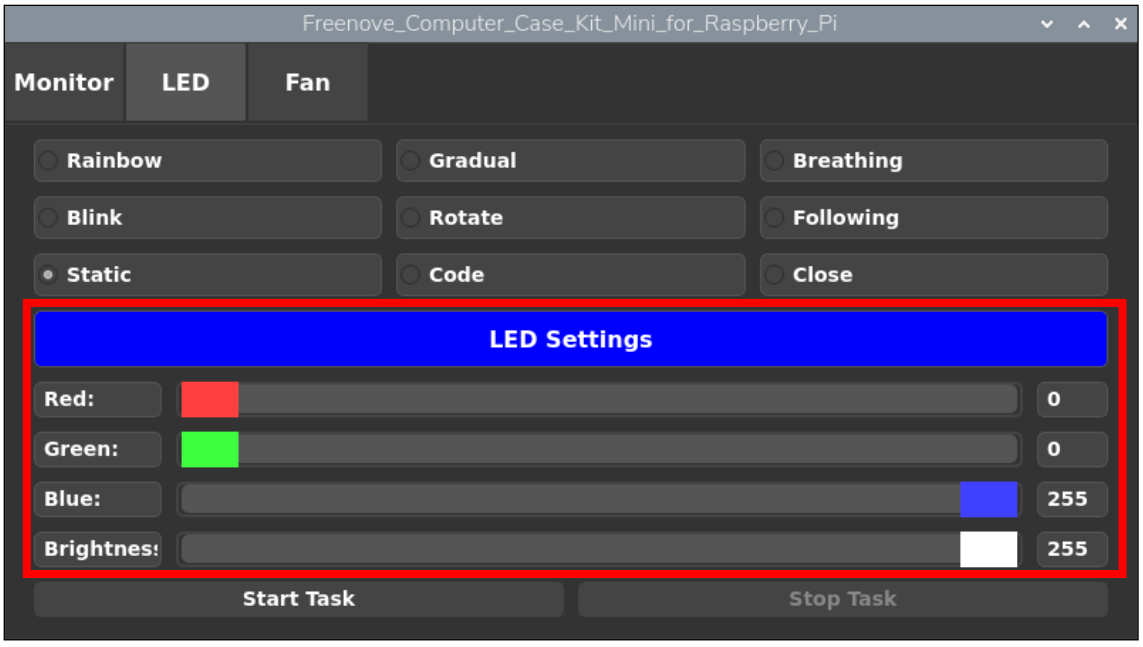
3.2.2 LED Control Interface

This is the control interface for the case ARGB lights. You can select different modes to display various lighting effects.

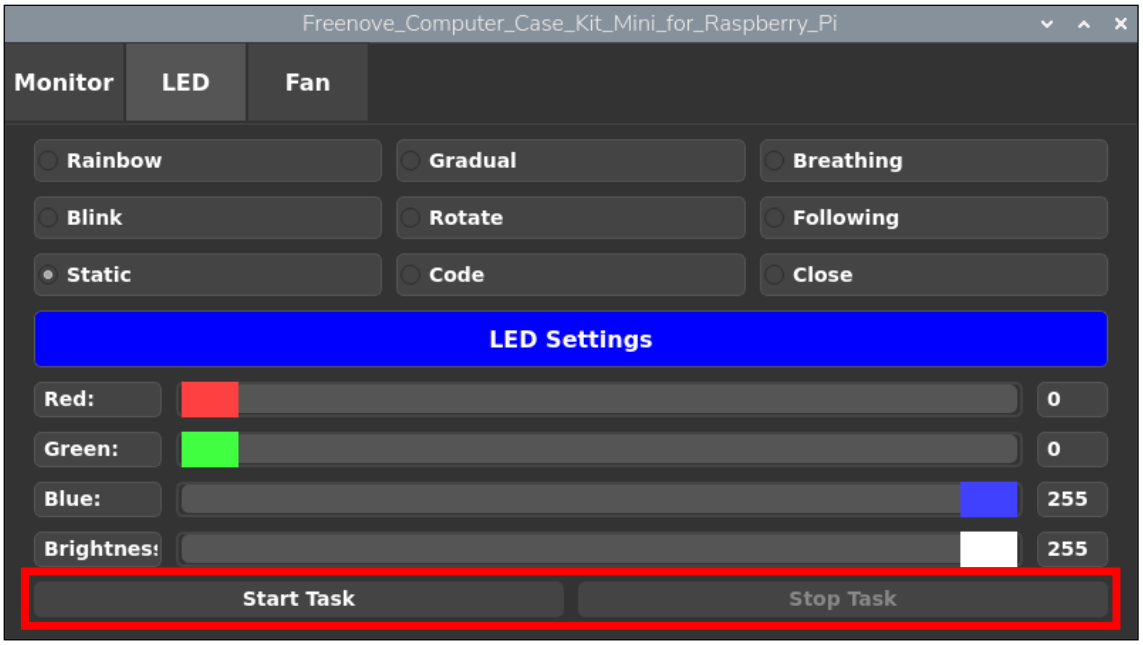


There are various preset lighting effect available, namely **Rainbow, Gradual, Breathing, Blink, Rotate, Following, Static and Code.** You can select the corresponding option to switch among the modes.

Note: Only in **Breathing, Blink, Rotate, Following, Static** modes can the color and brightness of the RGB lights be controlled using the slider below. In other modes, the color and brightness of the RGB lights cannot be adjusted.



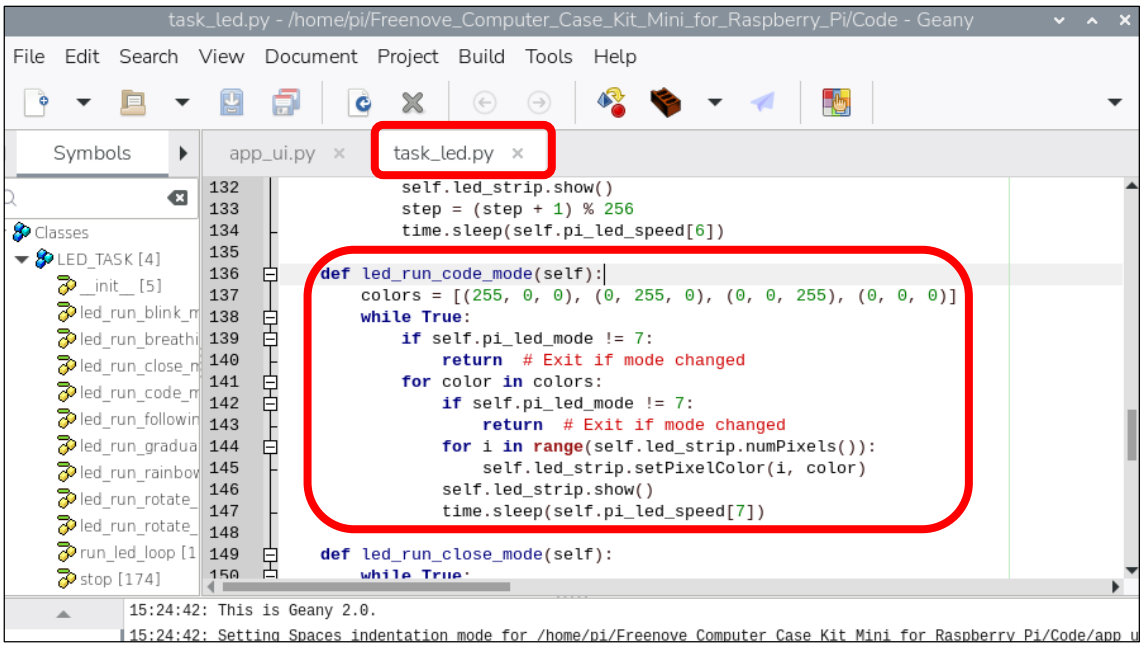
After selecting your preferred lighting effect, click the "**Start Task**" button. Both fans will then run with the chosen effect. To stop the lighting effect, click the "**Stop Task**" button.



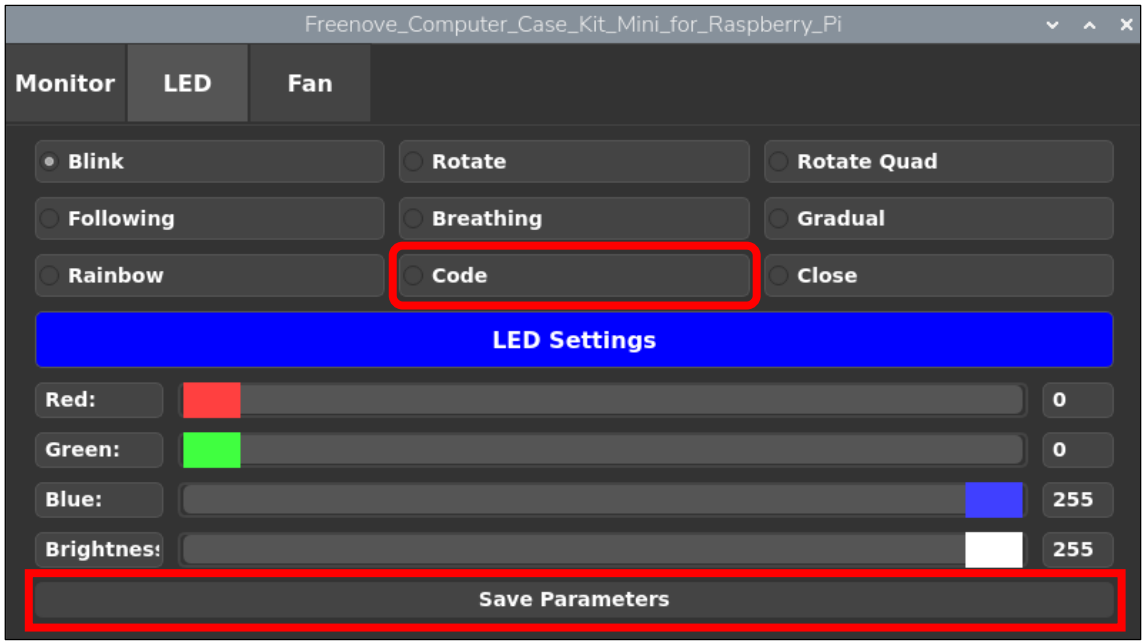
It is set to rainbow mode by default, as shown below.



If the preset mode does not meet your need, you can modify the **led_run_code_mode** function in the **task_led.py** file under the "Freenove_Computer_Case_Kit_Mini_for_Raspberry_Pi/Code" directory.



Restart the software, select "**Code**" mode, and then click "**Start Task**" to save it as default mode.



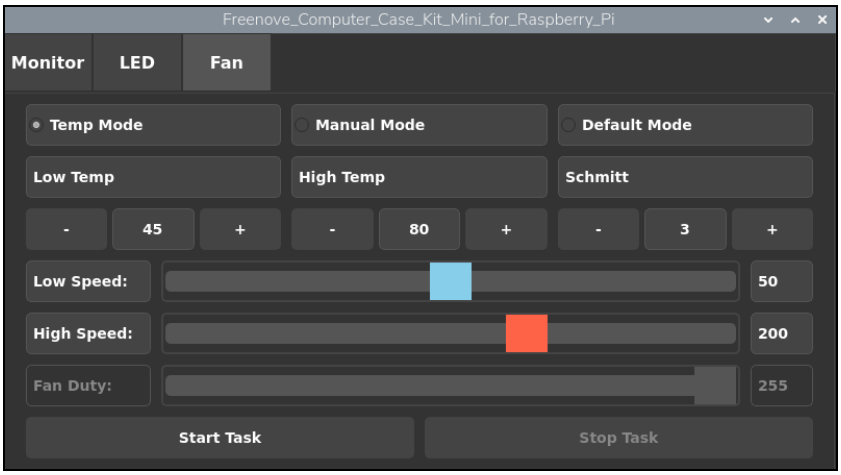
Important Notes:

1. Back up the source code before making any modifications.
2. Code mode is an advanced feature. Modifying the code in task_led.py requires a basic understanding of Python programming. If you are not familiar with programming, we do not recommend editing this file.
3. If, after modifying the code, you encounter either of the following issues: the software fails to start normally after a reboot, or the LED does not respond at all when clicking "Save Parameters," it indicates that an error in your code is preventing the program from running correctly. Please carefully review and correct the code.

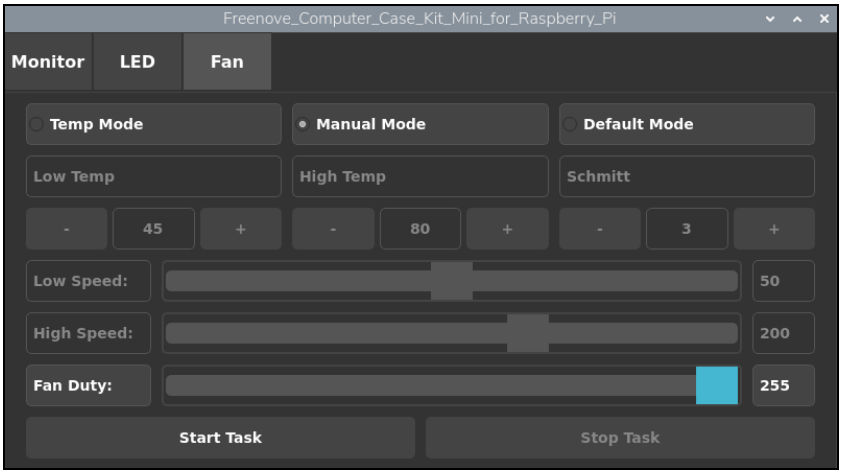
3.2.3 Fan Control Interface

This is the case fan control interface for convenient fan management. The interface will change when you select different modes: Temp mode, Manual mode, or Default mode.

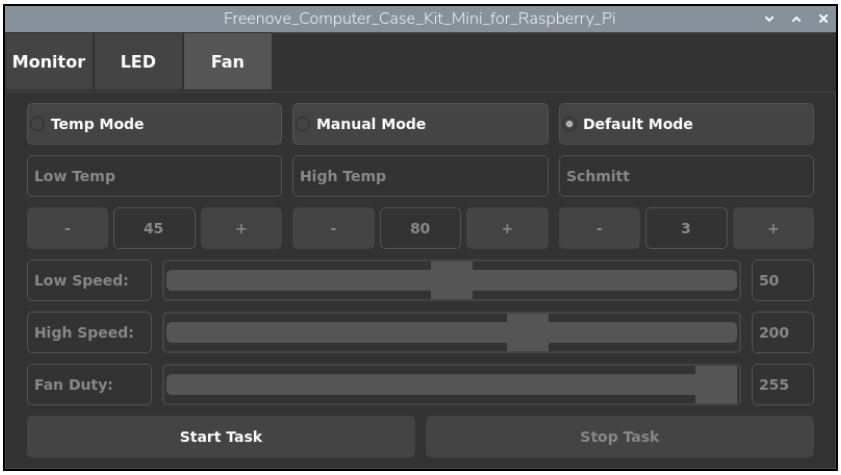
Temp Mode



Manual mode



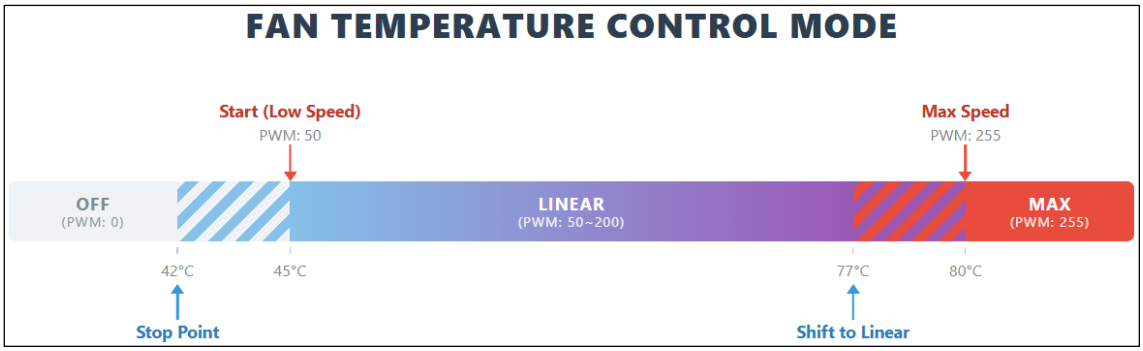
Default mode



Next, we will walk you through the different fan modes to help you better understand the case fan control interface.

Temp mode

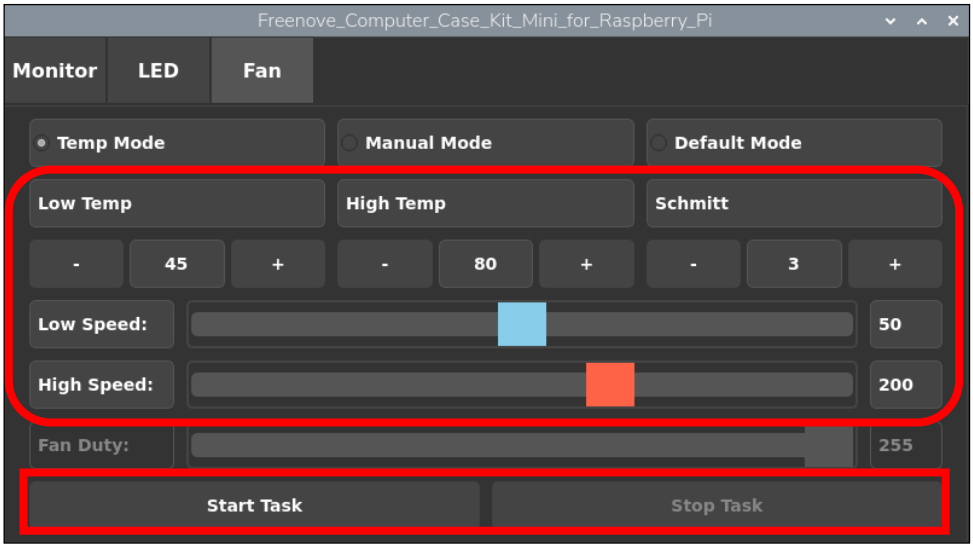
In Temp mode, the CPU temperature of the Raspberry Pi is sampled at regular intervals. The control logic for this process is shown in the following figure.



The diagram uses various colors to represent different temperature zones. All values shown are configurable defaults.

- **Heating Phase:**
  - When temperature ≥ **Low Temp** (default: 45°C), the fans start at **Low Speed** (default: 50).
  - When the temperature is within the set range (default: 45°C to 80°C), the fan's PWM value varies linearly within the configured range (default: 50 to 200). The higher the temperature, the faster the fan speed.
  - When temperature ≥ **High Temp** (default: 80°C), the fan speed increases to **full speed** (255).
- **Cooling Phase:**
  - When the temperature < **High Temp - Schmitt** (default: 80 - 3 = 77°C), the fan's PWM value varies linearly within the configured range (default: 50 to 200), ensuring a smooth and stable cooling process.
  - When temperature < **Low Temp- Schmitt** (default: 45 - 3 = 42°C), the fan stops.

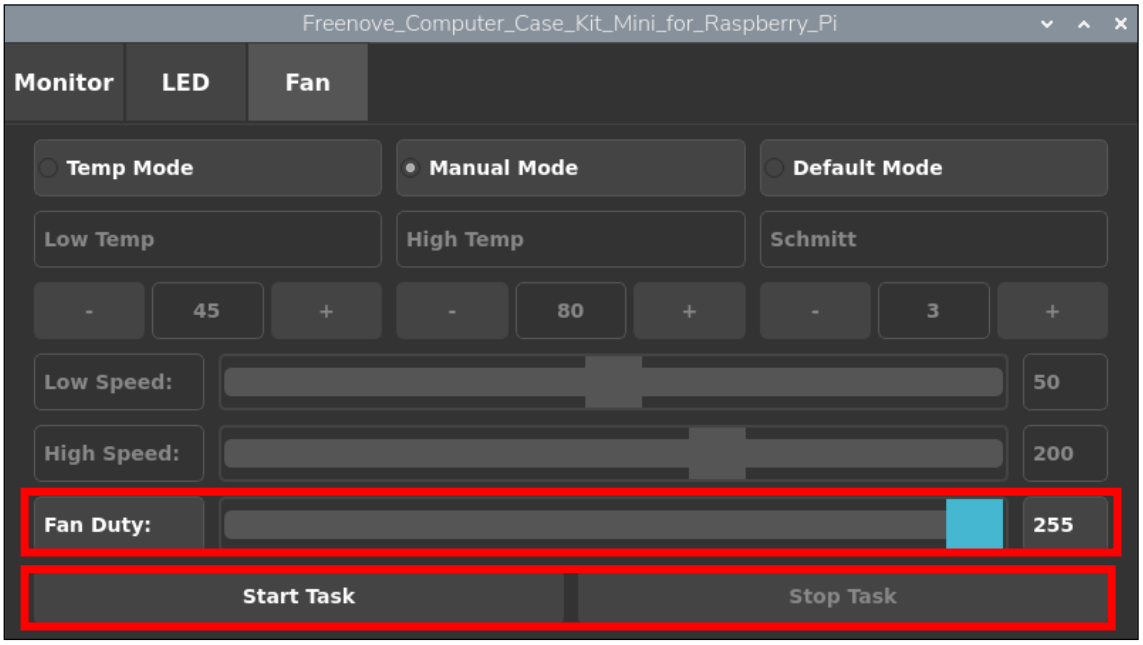
You can configure the following parameters via the software. After configuration, click the "**Start Task** " button to apply the settings. To turn off the fans, click the "**Stop Task**" button.



- **Low Temp**: The minimum temperature threshold to activate the case fans. Once reached, the fans begin spinning to accelerate airflow and expel warm air from the case.
- **High Temp**: A higher temperature threshold indicating significant heat buildup. When reached, the fans operate at a higher speed to enable rapid cooling of the system.
- **Schmitt (Hysteresis)**: A hysteresis value that prevents the fans from frequently switching on/off or changing speeds near a temperature threshold. It is recommended to keep it at the default value.
- **Low Speed**: The PWM duty cycle applied when the temperature first exceeds the Low Temp threshold. This low speed provides a quiet and power-efficient means of initial cooling.
- **High Speed**: The high PWM duty cycle activated when the temperature exceeds the High Temp threshold. This is critical during heavy computational loads to maximize cooling efficiency, expel heat rapidly, and maintain optimal Raspberry Pi performance.

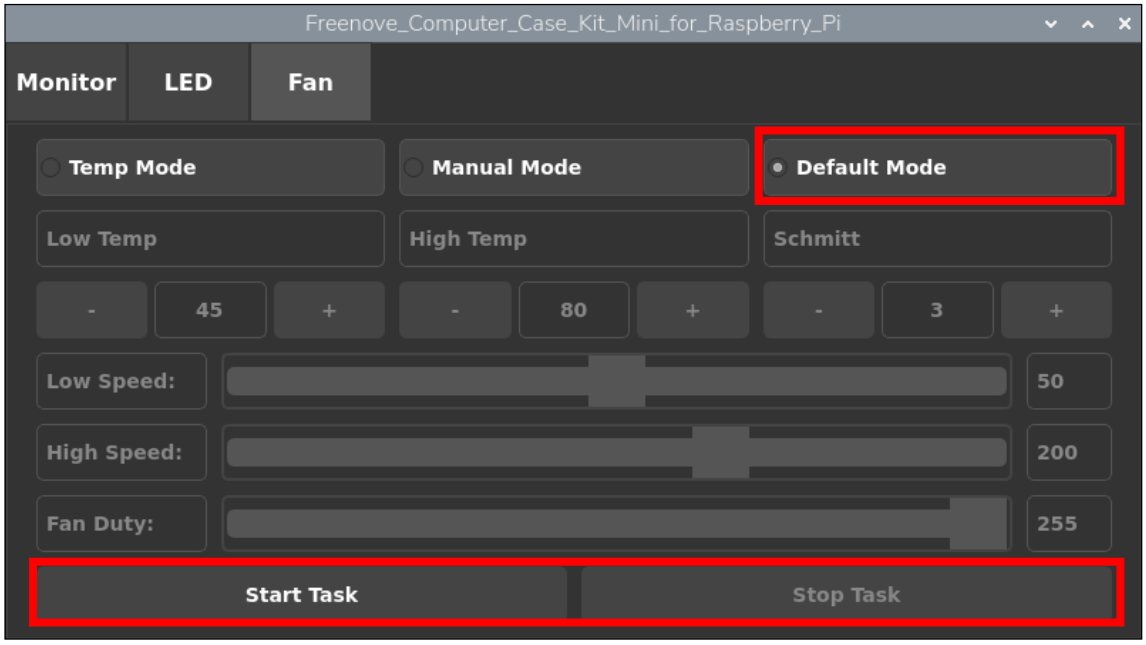
Manual mode

In this mode, you can manually adjust the fan's PWM value (range: 0–255) using the slider of "**Fan Duty**." After setting the desired value, click the "**Start Task**" button to apply the change. To turn off the fans, click the "**Stop Task**" button.



Defualt mode

Select "**Default Mode**" and click "**Start Task**" to apply the configuration. The fan will operate according to the default thermal management policy of the Raspberry Pi 5. To turn off the fans, click the "**Stop Task**" button.



As the temperature of the Raspberry Pi 5 increases, the fan reacts in the following way:

- below 50°C, the fan does not spin at all (0% speed).
- at 50°C, the fan turns on at a low speed (30% speed)
- at 60°C, the fan speed increases to a medium speed (50% speed)
- at 67.5°C, the fan speed increases to a high speed (70% speed)
- at 75°C the fan increases to full speed (100% speed)

Temperature decreases use the same mapping with a 5°C hysteresis; fan speed decreases when the temperature drops to 5°C below each of the above thresholds.

To learn more about the Raspberry Pi 5's default thermal management policy, please refer to the link below:
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#fan-cases

3.2.4 Disable Terminal Display & Disable sudo Password

If you do not need to view the debugging information and want to close the command terminal window, follow the steps below.

Open the terminal on your Raspberry Pi.

Run the following command to set the correct permissions for the Freenove desktop software:

```
ls -al ~/Desktop/FNK108.desktop
sudo chmod 777 ~/Desktop/ FNK108.desktop
```

```
pi@raspberrypi:~ $ ls -al ~/Desktop/FNK0108.desktop
-rwxr-xr-x 1 root root 255 May 22 08:54 /home/pi/Desktop/FNK0108.desktop
pi@raspberrypi:~ $ sudo chmod 777 ~/Desktop/FNK0108.desktop
pi@raspberrypi:~ $ ls -al ~/Desktop/FNK0108.desktop
-rwxrwxrwx 1 root root 255 May 22 08:54 /home/pi/Desktop/FNK0108.desktop
```

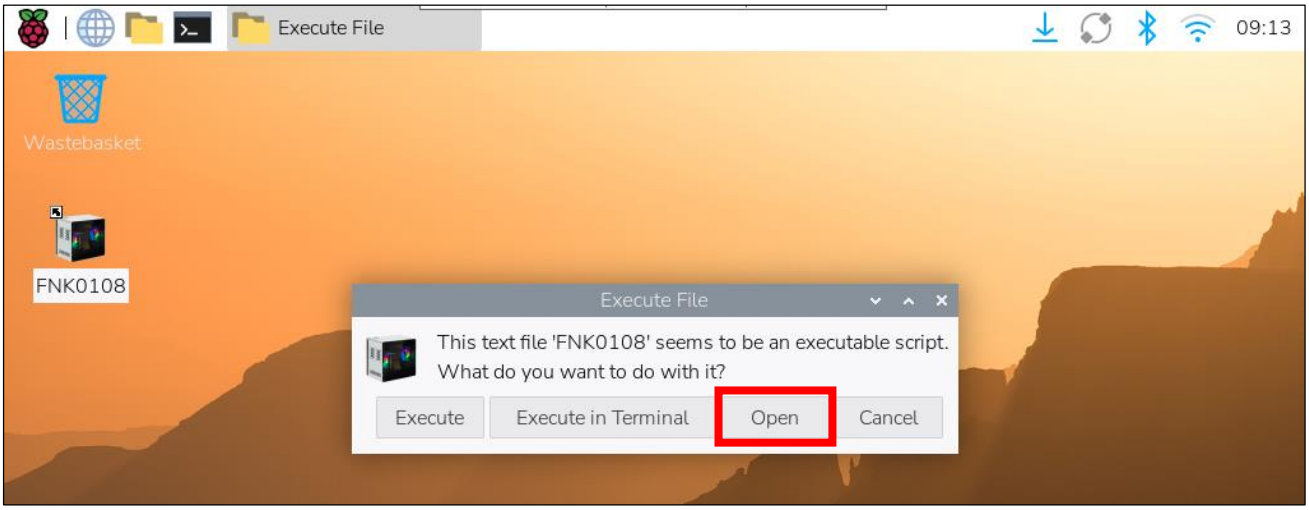
To revert to the default permissions, run the following command:

```
ls -al ~/Desktop/ FNK108.desktop
sudo chmod 755 ~/Desktop/FNK108.desktop
```

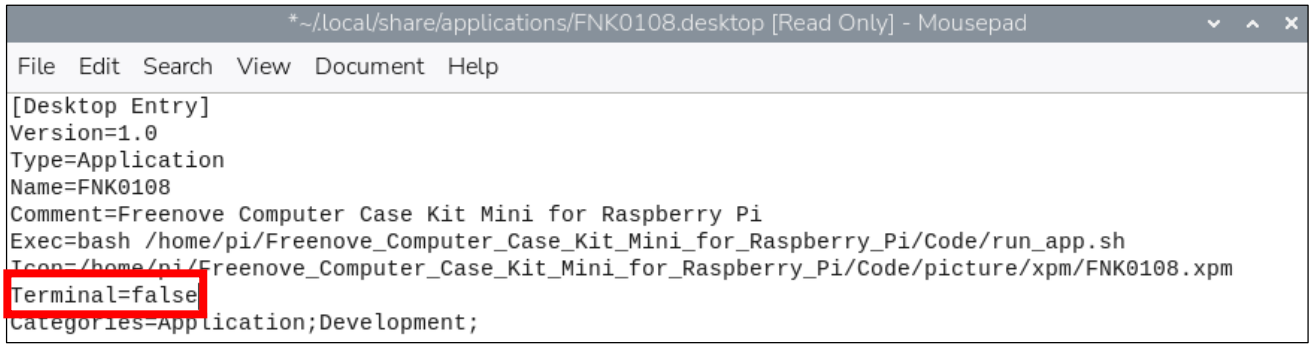
```
pi@raspberrypi:~ $ ls -al ~/Desktop/FNK0108.desktop
-rwxrwxrwx 1 root root 255 May 22 08:54 /home/pi/Desktop/FNK0108.desktop
pi@raspberrypi:~ $ sudo chmod 755 ~/Desktop/FNK0108.desktop
pi@raspberrypi:~ $ ls -al ~/Desktop/FNK0108.desktop
-rwxr-xr-x 1 root root 255 May 22 08:54 /home/pi/Desktop/FNK0108.desktop
```

After modifying the permissions for Freenove.desktop, please follow these steps:

1. Close the current upper-computer software completely.
2. Double-click the Freenove software on the desktop to restart the application.
3. In the pop-up window, click **Open** to launch the software.



Change the parameter of **Terminal** from true to **false**, save and clost the file.



Chapter 4 Speed Test

This is an additional chapter for users who wish to test the read and write speeds of their SSD and SD Card.

4.1 SD Card Speed Test

Raspberry Pi's official microSD cards support DDR50 and SDR104 bus speeds, with a maximum transfer rate of up to 104 MB/s.
https://www.raspberrypi.com/documentation/accessories/sd-cards.html#content
https://www.sdcard.org/developers/sd-standard-overview/bus-speed-default-speed-high-speed-uhs-sd-express/

Open the terminal and enter the following command:

```
git clone https://github.com/TheRemote/PiBenchmarks
```

```
pi@raspberrypi:~ $ git clone https://github.com/TheRemote/PiBenchmarks
Cloning into 'PiBenchmarks'...
remote: Enumerating objects: 652, done.
remote: Counting objects: 100% (112/112), done.
remote: Compressing objects: 100% (80/80), done.
remote: Total 652 (delta 69), reused 65 (delta 31), pack-reused 540 (from 1)
Receiving objects: 100% (652/652), 119.41 KiB | 269.00 KiB/s, done.
Resolving deltas: 100% (386/386), done.
```

Enter the directory:

```
cd PiBenchmarks/
```

Grant executable permissions to the script:

```
chmod +x Storage.sh
```

```
pi@raspberrypi:~ $ cd PiBenchmarks/
pi@raspberrypi:~/PiBenchmarks $ ls
CODE_OF_CONDUCT.md  CONTRIBUTING.md  LICENSE  README.md  Storage.sh
pi@raspberrypi:~/PiBenchmarks $ chmod +x Storage.sh
pi@raspberrypi:~/PiBenchmarks $
```

Start the speed test. Please be aware that the first execution will involve downloading the required dependencies, so the process could take a relatively long time.

```
sudo ./Storage.sh ~/
```

```
pi@raspberrypi:~/PiBenchmarks $ sudo ./Storage.sh ~/
Trimming and syncing drives ...
/boot/firmware: 424.5 MiB (445118464 bytes) trimmed on /dev/mmcblk0p1
/: 21.4 GiB (22927368192 bytes) trimmed on /dev/mmcblk0p2
Board information: Manufacturer: Raspberry Pi Foundation - Model: Raspberry Pi 5 Model B Rev 1.0 - Architecture:
 aarch64 - OS: Debian GNU/Linux 13 (trixie)
Fetching required components ...
Install lshw
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
Hit:4 http://archive.raspberrypi.com/debian trixie InRelease
Reading package lists... Done
```

After the speed test is completed, follow the prompts to enter a description and a name for your SSD (you can use any arbitrary characters).

```
RandRead: 8181 - RandWrite: 2405 - Read: 13783 - Write: 2344
Enter a description of your storage and setup (Example: Kingston A400 SSD on Pi 4 using StarTech SATA to USB ada
pter)
Description: sd1
(Optional) Enter alias to use on benchmark results.  Leave blank for completely anonymous.
Alias (leave blank for Anonymous): sd1
Result submitted successfully and will appear live on https://pibenchmarks.com within a couple of minutes.
```

Test result:

Category	Test	Result
HDParm	Disk Read	87.49 MB/sec
HDParm	Cached Disk Read	88.76 MB/sec
DD	Disk Write	21.1 MB/s
FIO	4k random read	2261 IOPS (9047 KB/s)
FIO	4k random write	598 IOPS (2395 KB/s)
IOZone	4k read	13783 KB/s
IOZone	4k write	2344 KB/s
IOZone	4k random read	8181 KB/s
IOZone	4k random write	2405 KB/s
	Score: 1095	

4.2 SSD Speed Test

The Raspberry Pi 5 includes a PCIe x1 slot that is certified for PCIe Gen 2.0, providing a theoretical maximum throughput of 5GT/sec, which roughly translates to 500MB/sec for read and write operations. Although this slot is not officially certified for PCIe Gen 3.0, it is possible to force the use of Gen 3.0 for potentially higher speeds. **At the time of this writing, due to the limitation of the expansion chip, both the Dual-NVMe Adapter Board and Quad-NVMe Adapter Board only support the PCIe 2.0 protocol.**

In actual tests, it is found that most SSDs can work stably at PCIE2.0, but are slightly unstable at PCIE3.0, while others are just the opposite. Therefore, please choose PCIE2.0 or PCIE3.0 according to your actual situation.

The PCIe consortium states that the speed of PCIe Gen 3.0 x1 is up to 8GT/sec, which translates to approximately 985MB/sec.
https://en.wikipedia.org/wiki/PCI_Express#Comparison_table
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#pcie-gen-3-0

4.2.1 Mounting SSD Devices

Open the terminal, run the following commpad to get the SSD's information.

```
lsblk
```

```
pi@raspberrypi:~ $ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0         7:0    0     2G  0 loop
mmcblk0     179:0    0  29.3G  0 disk
├─mmcblk0p1 179:1    0   512M  0 part /boot/firmware
└─mmcblk0p2 179:2    0  28.8G  0 part /
zram0       254:0    0     2G  0 disk [SWAP]
nvme0n1     259:0    0 119.2G  0 disk
└─nvme0n1p1 259:4    0 119.2G  0 part
nvme1n1     259:1    0 119.2G  0 disk
└─nvme1n1p1 259:6    0 119.2G  0 part
nvme2n1     259:2    0 119.2G  0 disk
└─nvme2n1p1 259:7    0 119.2G  0 part
nvme3n1     259:3    0 119.2G  0 disk
└─nvme3n1p1 259:5    0 119.2G  0 part
```

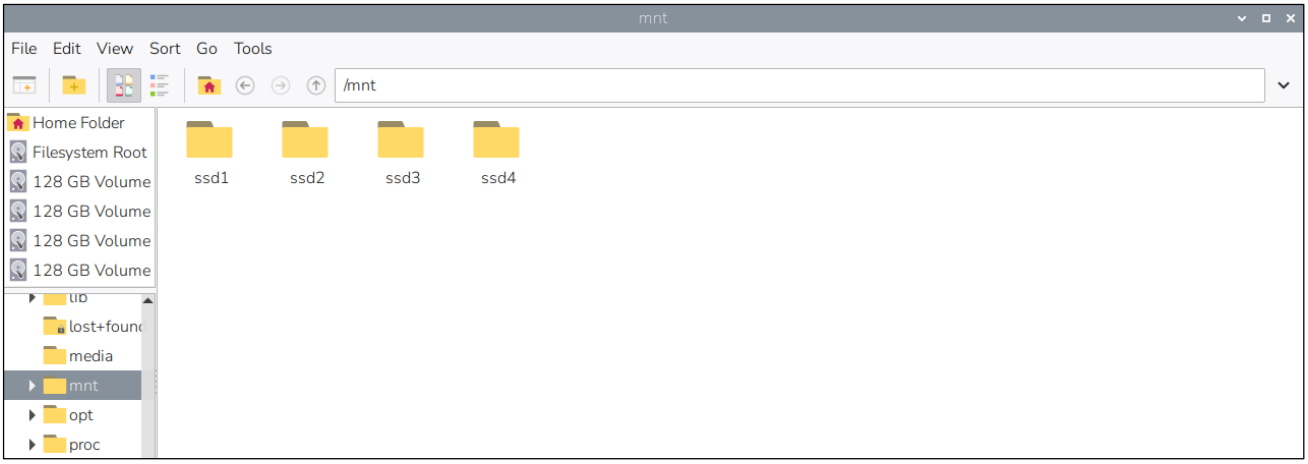
As shown in the image above, the Raspberry Pi 5 has successfully detected four NVMe SSDs, with mmcb0p1 being the system drive.

Prior to this, we have already partitioned all NVMe SSD devices (**If your SSD is not partitioned, please refer to Section 2.3 in the "Installing Raspberry Pi OS.pdf" file**). Next, we need to mount all the SSDs so they can be used by the Raspberry Pi's file system. If you have already mounted them, click here to proceed to the next step.

```
sudo mkdir -p /mnt/ssd1
sudo mkdir -p /mnt/ssd2
sudo mkdir -p /mnt/ssd3
sudo mkdir -p /mnt/ssd4
```

```
pi@raspberrypi:~ $ sudo mkdir -p /mnt/ssd1
pi@raspberrypi:~ $ sudo mkdir -p /mnt/ssd2
pi@raspberrypi:~ $ sudo mkdir -p /mnt/ssd3
pi@raspberrypi:~ $ sudo mkdir -p /mnt/ssd4
```

You can view the created SSD mount points (ssd1/ssd2/ssd3/ssd4) in the /mnt directory.



Execute the following commands to complete the mounting process.

```
sudo mount /dev/nvme0n1p1 /mnt/ssd1
sudo mount /dev/nvme1n1p1 /mnt/ssd2
sudo mount /dev/nvme2n1p1 /mnt/ssd3
sudo mount /dev/nvme3n1p1 /mnt/ssd4
```

```
pi@raspberrypi:~ $ sudo mount /dev/nvme0n1p1 /mnt/ssd1
pi@raspberrypi:~ $ sudo mount /dev/nvme1n1p1 /mnt/ssd2
pi@raspberrypi:~ $ sudo mount /dev/nvme2n1p1 /mnt/ssd3
pi@raspberrypi:~ $ sudo mount /dev/nvme3n1p1 /mnt/ssd4
```

4.2.2 Speed Test

After completing the mounting of all NVMe SSD devices, execute the following command to enter the PiBenchmarks directory. We will proceed with individual device speed testing.

```
cd PiBenchmarks/
```

```
pi@raspberrypi:~ $ cd PiBenchmarks/
pi@raspberrypi:~/PiBenchmarks $
```

Run the following command to start the speed test for the first mounted device, **ssd1**.

```
sudo ./Storage.sh /mnt/ssd1
```

```
pi@raspberrypi:~/PiBenchmarks $ sudo ./Storage.sh /mnt/ssd1
Trimming and syncing drives ...
/mnt/ssd4: 116.8 GiB (125423808512 bytes) trimmed on /dev/nvme3n1p1
/mnt/ssd3: 116.8 GiB (125423808512 bytes) trimmed on /dev/nvme2n1p1
/mnt/ssd2: 116.8 GiB (125423808512 bytes) trimmed on /dev/nvme1n1p1
/mnt/ssd1: 116.8 GiB (125423808512 bytes) trimmed on /dev/nvme0n1p1
/boot/firmware: 424.5 MiB (445118464 bytes) trimmed on /dev/mmcblk0p1
/: 21.3 GiB (22861807616 bytes) trimmed on /dev/mmcblk0p2
Board information: Manufacturer: Raspberry Pi Foundation - Model: Raspberry Pi 5
 Model B Rev 1.0 - Architecture: aarch64 - OS: Debian GNU/Linux 13 (trixie)
Fetching required components ...
Clock speeds: CPU: 2400 - Core: 910
Chosen partition (/mnt/ssd1) has been detected as /dev/nvme0n1p1 (nvme0n1p1)
Starting INXI hardware identification...
```

After the speed test is completed, follow the prompts to enter a description and a name for your SSD (you can use any arbitrary characters).

```
RandRead: 46786 - RandWrite: 95069 - Read: 125241 - Write: 96006
Enter a description of your storage and setup (Example: Kingston A400 SSD on Pi 4 using
StarTech SATA to USB adapter)
Description: ssd1
(Optional) Enter alias to use on benchmark results.  Leave blank for completely anonymou
s.
Alias (leave blank for Anonymous): ssd1
Result submitted successfully and will appear live on https://pibenchmarks.com within a
couple of minutes.
```

Test result:

Category	Test	Result
HDParm	Disk Read	387.83 MB/sec
HDParm	Cached Disk Read	376.84 MB/sec
DD	Disk Write	339 MB/s
FIO	4k random read	95255 IOPS (381023 KB/s)
FIO	4k random write	76704 IOPS (306816 KB/s)
IOZone	4k read	170906 KB/s
IOZone	4k write	135278 KB/s
IOZone	4k random read	51424 KB/s
IOZone	4k random write	137275 KB/s

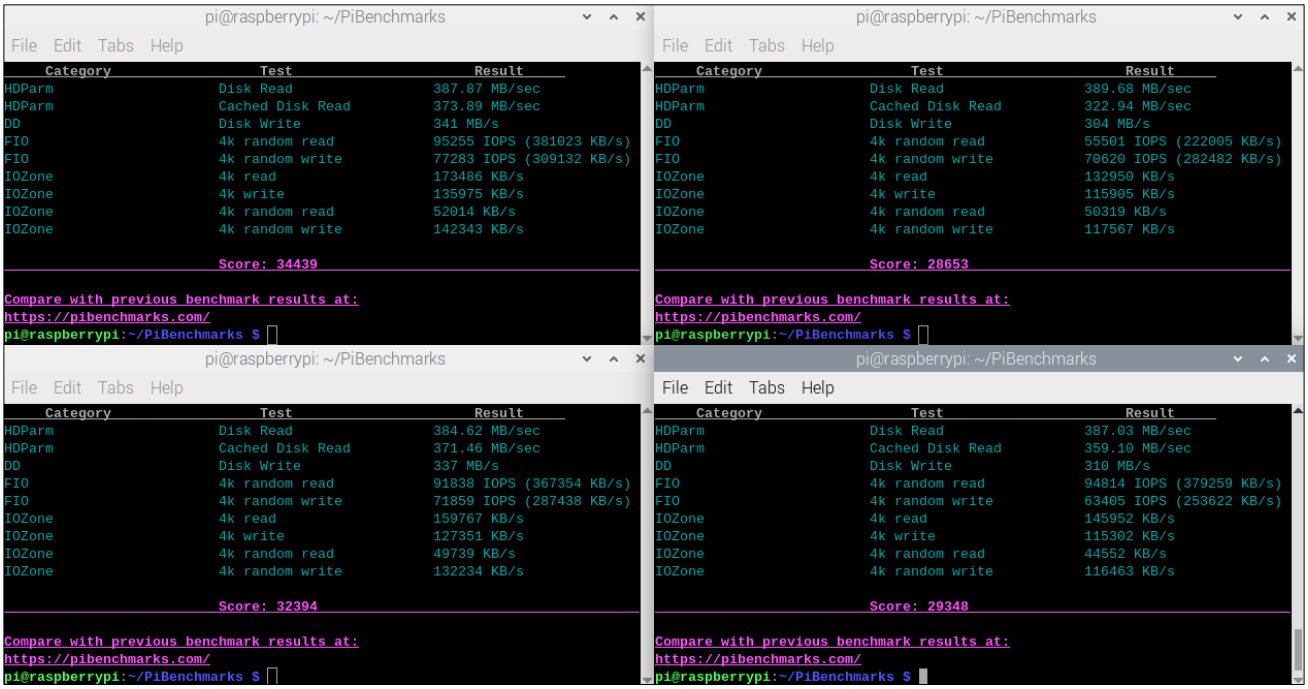
Score: 33798

Next, we will sequentially execute the following commands to perform individual speed tests on each SSD.

```
sudo ./Storage.sh /mnt/ssd2
sudo ./Storage.sh /mnt/ssd3
sudo ./Storage.sh /mnt/ssd4
```

```
pi@raspberrypi:~/PiBenchmarks $ sudo ./Storage.sh /mnt/ssd2
Trimming and syncing drives ...
/mnt/ssd4: 0 B (0 bytes) trimmed on /dev/nvme3n1p1
/mnt/ssd3: 0 B (0 bytes) trimmed on /dev/nvme2n1p1
/mnt/ssd2: 0 B (0 bytes) trimmed on /dev/nvme1n1p1
/mnt/ssd1: 624.2 MiB (654471168 bytes) trimmed on /dev/nvme0n1p1
/boot/firmware: 424.5 MiB (445118464 bytes) trimmed on /dev/mmcblk0p1
/: 0 B (0 bytes) trimmed on /dev/mmcblk0p2
Board information: Manufacturer: Raspberry Pi Foundation - Model: Raspberry Pi 5 Model B
```

The performance varies among different SSDs, and each test may have certain error, which is normal. The following figures show the speed test results of the four SSDs.



4.2.3 PCIe Gen3.0 Speed Test

In the Preface, it is mentioned that the Raspberry Pi's PCIe Gen 3.0 has not been officially certified. While it is functional, its performance is not as reliable as desired. This chapter is presented as an exploratory section for assessing the speed capabilities of SSDs when used with PCIe Gen 3.0. **For practical applications, it is advised to opt for PCIe Gen 2.0 to ensure greater stability and dependability.**

If your kit includes the single-slot NVMe Adapter Board (Model: **FNK0108A/H)**, you can enable PCIe Gen 3.0 mode. If your adapter is a different model or no adjustment is needed, please skip this step.

Enable PCIe Gen 3.0

Add the line `dtparam=pciex1_gen=3` to /boot/firmware/config.txt to enable PCIe Gen3.0.

As shown below, enter the command to open the file.

```
sudo nano /boot/firmware/config.txt
```

Add the line `dtparam=pciex1_gen=3` to the end of the file, as shown below:

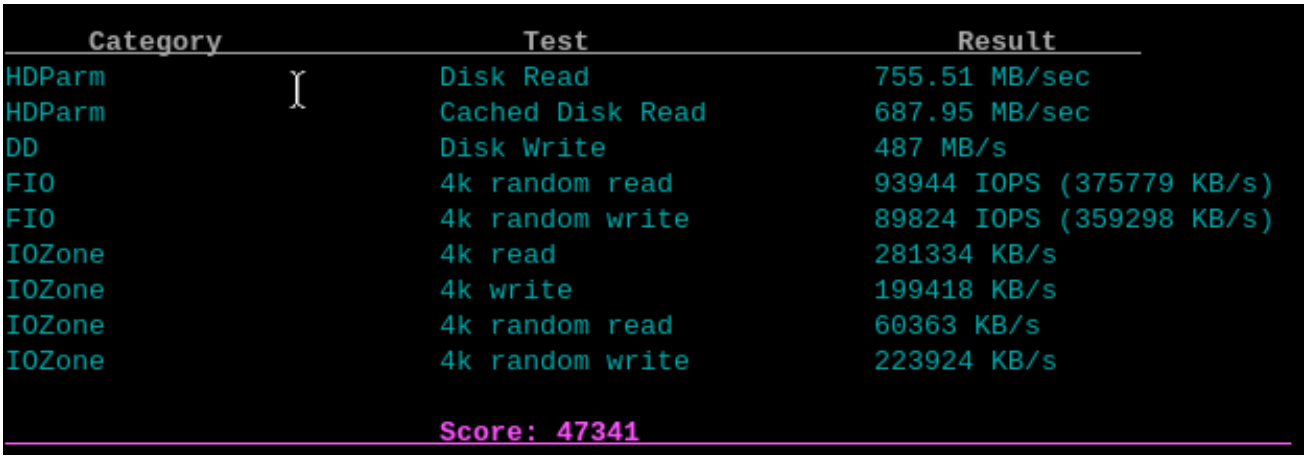


Press Ctrl-O to save the file, Enter to confirm, and Ctrl-X to exit.

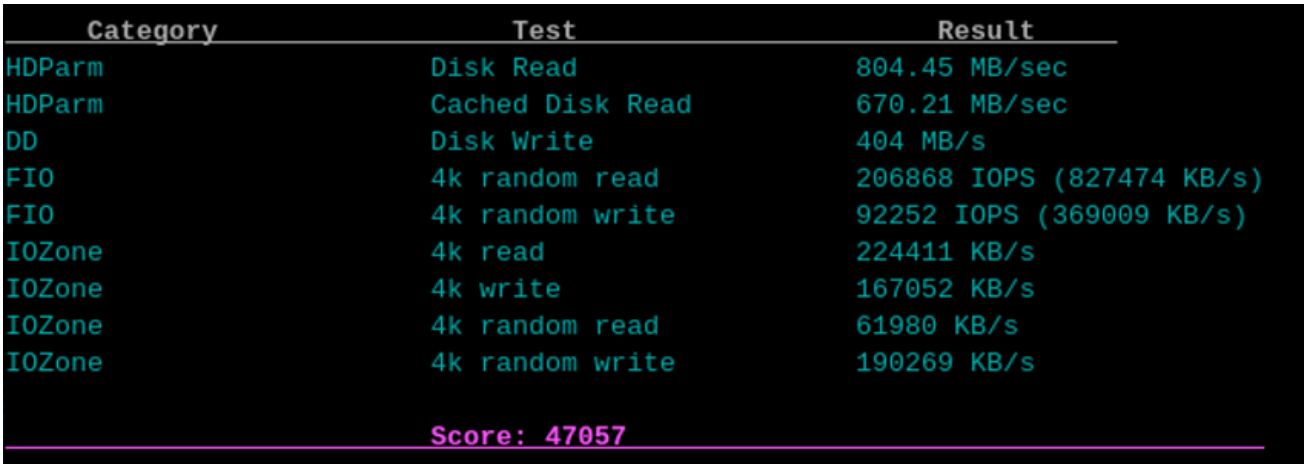
Reboot your Raspberry Pi.

```
sudo reboot
```

After rebooting, test the speed again.



The speed of another SSD.



Category	Test	Result
HDParm	Disk Read	804.45 MB/sec
HDParm	Cached Disk Read	670.21 MB/sec
DD	Disk Write	404 MB/s
FIO	4k random read	206868 IOPS (827474 KB/s)
FIO	4k random write	92252 IOPS (369009 KB/s)
IOZone	4k read	224411 KB/s
IOZone	4k write	167052 KB/s
IOZone	4k random read	61980 KB/s
IOZone	4k random write	190269 KB/s
	Score: 47057	

Disable PCIe Gen 3.0

Delete the line added with the previous step to disable PCIe Gen3.0.

Delete the line `dtparam=pciex1_gen=3` in the boot/firmware/config.txt file.

After the line is removed, it will change to PCIe Gen2.0.

What's next?

Thank you again for choosing Freenove products.

THANK YOU for participating in this learning experience!

We have reached the end of this Tutorial. If you find errors, omissions, or you have suggestions and/or questions about the Tutorial or component contents of this Kit, please feel free to contact us:
support@freenove.com

We will make every effort to make changes and correct errors as soon as feasibly possible and publish a revised version.

If you want to learn more about Arduino, ESP32, Raspberry Pi Pico W, Raspberry Pi, Smart Cars, Robotics and other interesting products in science and technology, please continue to visit our website. We will continue to launch fun, cost effective, innovative and exciting products.
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